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Analysis of Food Security and Nutritional Status in Ghana

Master's Thesis

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ABSTRACT

This thesis investigates food security and nutritional status in Ghana, focusing on four main objectives: assessing current food security trends, identifying key influencing factors, evaluating policy interventions, and examining progress toward Sustainable Development Goals (SDGs) 1, 2, and 3. The study employed secondary data analysis using national surveys and international databases, including the Global Food Security Index (GFSI). Descriptive and trend analyses were conducted across key food security dimensions: availability, affordability, quality, and sustainability. The analysis shows modest progress in food availability, with Ghana's farm infrastructure score improving by 48.2 points, and volatility in agricultural production remaining low (score: 91.8). However, affordability dropped by 4.4 points, and food prices surged by over 30%, contributing to food insecurity among 49.1% of Ghanaians. Nutritional outcomes remain concerning, with 66% of children under five affected by anemia, and stunting prevalent in 17.5% of children. The absence of a national nutrition plan led to a drop in nutritional standards from 57.1 to 29.8 points. On sustainability, Ghana scored poorly at 45.1, reflecting weak disaster preparedness and adaptation capacity. Government efforts, such as the Planting for Food and Jobs initiative, improved supply sufficiency, yet gaps in safety nets and regional disparities persist. Food insecurity affects 46.7% in the north compared to 13.8% in the south. Although Ghana ranks 3rd in food security in Sub-Saharan Africa, it is still off-track for achieving SDGs 1, 2, and 3 by 2030. The study recommends enhancing post-harvest infrastructure, implementing a national nutrition policy, scaling social protection, and adopting climate-smart agricultural strategies.

Keywords: food security, nutritional status, Ghana, GFSI, policy effectiveness, SDGs

ABSTRAKTNÍ

Tato práce zkoumá bezpečnost potravin a nutriční status v Ghaně se zaměřením na čtyři hlavní cíle: hodnocení současných trendů na bezpečnost potravin, identifikace klíčových ovlivňujících faktorů, hodnocení politických zásahů a zkoumání pokroku směrem k udržitelným rozvojovým cílům (SDG) 1, 2 a 3. Popisné a trendové analýzy byly provedeny napříč klíčovými dimenzemi zabezpečení potravin: dostupnost, dostupnost, kvalita a udržitelnost. Analýza ukazuje skromný pokrok v dostupnosti potravin, přičemž ghanská zemědělská infrastruktura se zlepšuje o 48,2 bodů a volatilita v zemědělské produkci zůstává nízké (skóre: 91,8). Dostupnost však klesla o 4,4 bodů a ceny potravin vzrostly o více než 30%, což přispělo k nejistotě potravin mezi 49,1% Ghanaanů. Nutriční výsledky zůstávají týkající se, přičemž 66% dětí do pěti postižených anémií a zakrýváním převládajících u 17,5% dětí. Absence národního plánu výživy vedla k poklesu nutričních standardů z 57,1 na 29,8 bodů. Pokud jde o udržitelnost, Ghana skóroval špatně na 45,1, což odráželo slabou připravenost na katastrofy a adaptační kapacitu. Vládní úsilí, jako je iniciativa pro výsadbu potravin a pracovních míst, zlepšila dostatečnost nabídek, přesto přetrvávají mezery v bezpečnostních sítích a regionálních rozdílech. Nejistota potravin postihuje na severu 46,7% ve srovnání s 13,8% na jihu. Ačkoli Ghana se řadí na 3. místo v oblasti potravinářské bezpečnosti v subsaharské Africe, je stále mimo trať pro dosažení SDG 1, 2 a 3 do roku 2030. Studie doporučuje posílit po sklizni infrastrukturu, provádět národní nutriční politiku, škálování sociální ochrany a přijímat klimatizované zemědělské strategie.

Klíčová slova: Zabezpečení potravin, stav výživy, Ghana, GFSI, Efektivita politiky, SDG

Table of Contents

1	INTRODUCTION.....	1
1.1.	<i>Study Background.....</i>	<i>1</i>
1.2.	<i>Problem Statement</i>	<i>2</i>
1.3.	<i>Significance of Study.....</i>	<i>3</i>
2	AIM AND METHODOLOGY.....	4
2.1.	<i>Aim of the Study</i>	<i>4</i>
2.2.	<i>Research Questions to be Answered</i>	<i>4</i>
2.3.	<i>Research Methods</i>	<i>4</i>
2.3.1.	<i>Sources of Data</i>	<i>5</i>
2.3.2.	<i>Data Analysis Procedures</i>	<i>5</i>
2.3.3.	<i>Limitations of the Study.....</i>	<i>5</i>
2.4.	<i>Structure of the Work.....</i>	<i>6</i>
3	LITERATURE REVIEW	7
3.1.	<i>Review of Relevant Theories.....</i>	<i>7</i>
3.1.1	<i>Entitlement Theory.....</i>	<i>7</i>
3.1.2	<i>Food System Approach</i>	<i>8</i>
3.1.3	<i>Malthusianism.....</i>	<i>9</i>
3.2.	<i>The Concept of Nutritional Status.....</i>	<i>11</i>
3.2.1.	<i>Dietary Intake.....</i>	<i>11</i>
3.2.3.	<i>Physiological and Metabolic Functions</i>	<i>12</i>
3.3.	<i>The Concept of Food Security</i>	<i>12</i>
3.4.	<i>Dimensions of Food Security.....</i>	<i>13</i>
3.4.1.	<i>Availability</i>	<i>14</i>
3.4.2.	<i>Access</i>	<i>15</i>
3.4.3.	<i>Utilization</i>	<i>18</i>
3.4.4	<i>Stability.....</i>	<i>19</i>

3.5.	<i>Levels of Food Security Measurement</i>	20
3.6.	<i>Food Insecurity.....</i>	22
3.7.	<i>Food Insecurity in Africa.....</i>	23
3.8.	<i>Food Security and SDG</i>	24
3.9.	<i>Food Insecurity in Ghana.....</i>	25
3.10.	<i>Nutrition Based Diseases and its Prevalence in Ghana</i>	26
RESULTS AND DISCUSSION.....		27
4.1.	<i>Country Profile.....</i>	27
4.2.	<i>The Current Trends and Status of Food Security in Ghana</i>	27
4.2.1	<i>Food Security Environment.....</i>	28
4.2.2	<i>Affordability</i>	30
4.2.3	<i>Availability</i>	32
4.2.4	<i>Quality and Safety</i>	34
4.2.5	<i>Sustainability and Adaptation.....</i>	36
4.3.	<i>Key Factors Influencing Food Security and Nutritional Status in Ghana.....</i>	37
4.3.1.	<i>Escalating Food Costs and Weakened Safety Nets.....</i>	37
4.3.2	<i>Poor Supply Chain and High Food Loss</i>	39
4.3.3	<i>Inadequate Nutritional Standards and Low Dietary Diversity</i>	40
4.3.4	<i>Weak Water and Disaster Risk Management.....</i>	41
4.4	<i>Existing Government Policies, Interventions, and Progress Made in Addressing Food Security Challenges in Ghana</i>	43
4.4.1	<i>Planting for Food and Jobs (PFJ)</i>	43
4.4.2	<i>Ghana School Feeding Programme (GSFP)</i>	44
4.4.3	<i>One Village, One Dam Initiative</i>	44
4.4.4	<i>National Food Buffer Stock Company (NAFCO).....</i>	45
4.5	<i>Ghana's Progress Towards Achieving SDG 1 (No Poverty), SDG 2 (Zero Hunger), and SDG 3 (Good Health and Well-being) by 2030.....</i>	46
4.5.1	<i>SDG 1: No Poverty</i>	47
4.5.2	<i>SDG 2: Zero Hunger</i>	48

4.5.3	SDG 3: Good Health and Well-being.....	50
5	CONCLUSION AND RECOMMENDATIONS	53
5.1.	<i>Conclusions</i>	<i>53</i>
5.2.	<i>Recommendations</i>	<i>54</i>
	REFERENCES	55

List of Tables

Table 1: Selected countries' rank and score on food availability 2022.....	15
Table 2:Selected countries' rank and score on food access 2022.....	17
Table 3: Selected countries' rank and score on food utilization 2022	19
Table 4: Selected countries' rank and score on food stability 2022.....	20
Table 5: Sustainability and adaptation metrics for Ghana (2012 vs 2022).....	36
Table 6: selected crop production in Ghana from 2012 to 2021('000 metric tons)	43
Table 7: Ghana school feeding programme: impacts and challenges, 2018-2022	44
Table 8: one village, one dam implementation status (2023)	45
Table 9: SDGs 1;2;3 Progress.....	46

LIST OF FIGURES

Figure 1: Average overall score, global 2012-2022.....	13
Figure 2: Change in global average food costs,2012-2022.....	16
Figure 3: Ghana's food security index 2022	28
Figure 4: Ghana's food security ranking in Sub-Saharan Africa	29
Figure 5: Food security trends in selected countries from 2018 to 2022.....	30
Figure 6: Ghana's food affordability metrics (2012 vs 2022).....	31
Figure 7: Ghana's food availability metrics (2012 vs 2022).....	33
Figure 8: Ghana's food quality and safety metrics (2012 vs 2022)	35
Figure 9: Ghana's progress towards SDG 1	47
Figure 10: Ghana's progress towards SDG 2	49
Figure 11: Ghana's progress towards SDG 3	51

LIST OF ABBREVIATIONS

AU	African Union
CAADP	Comprehensive Africa Agriculture Development Programme
DHS	Demographic and Health Survey
FAO	Food and Agriculture Organization of the United Nations
FASDEP	Food and Agriculture Sector Development Policy
FCS	Food Consumption Score
GFSI	Global Food Security Index
GHS	Ghana Health Service / Ghanaian Cedi (context-dependent)
GIPC	Ghana Investment Promotion Centre
GSS	Ghana Statistical Service
HDDS	Household Dietary Diversity Score
HFIAS	Household Food Insecurity Access Scale
IFPRI	International Food Policy Research Institute
LEAP	Livelihood Empowerment Against Poverty
MoFA	Ministry of Food and Agriculture
NAFCO	National Food Buffer Stock Company
NCDs	Non-Communicable Diseases
SDGs	Sustainable Development Goals
SUN	Scaling Up Nutrition
UNICEF	United Nations Children's Fund
UNOCHA	United Nations Office for the Coordination of Humanitarian Affairs
USA	United States of America
WFP	World Food Programme
WHO	World Health Organization
WTO	World Trade Organization

1 INTRODUCTION

1.1. Study Background

Food security, according to the Food and Agriculture Organization (FAO), is a state in which individuals consistently have adequate physical, social, and economic access to safe, nutritious, and sufficient food that supports a healthy and active lifestyle (UNICEF, 2021). Despite notable improvements in agricultural productivity and food distribution, millions worldwide continue to experience hunger and malnutrition.

Sub-Saharan Africa faces some of the most severe food security challenges in the world, with approximately 256 million people experiencing undernourishment (Food and Agriculture Organization [FAO]., 2022). West Africa, in particular, struggles with climate variability, political instability, and rapid population growth that strain existing agricultural systems (Van Ittersum et al., 2016). The region's heavy dependence on rain-fed agriculture makes food production vulnerable to increasingly unpredictable weather patterns, while weak infrastructure limits market access and increases post-harvest losses (Sheahan & Barrett, 2017).

Ghana, once considered a success story in West African food security, has experienced fluctuating progress in recent years (Adom, 2018). Although the country has achieved lower-middle-income status and reduced extreme poverty from 52.6% in the early 1990s to approximately 23% today, significant regional disparities exist (World Bank, 2020). According to Nyantakyi-Frimpong and Bezner Kerr (2017), the northern regions of Ghana remain particularly vulnerable to food insecurity, with malnutrition rates substantially higher than the national average. Ghana's agricultural sector, which employs a significant portion of the population, is pivotal to the nation's food security (Darfour & Rosentrater, 2016). Approximately 39.74% of the total employment in Ghana was in agriculture in 2022. The sector contributed about GHS 90 billion to the economy in 2021, representing 21% of the overall economic output (Ghana Investment Promotion Center, 2022).

Despite advancements, the agricultural sector continues to face significant challenges, including insufficient investment in modern farming technologies, dependence on rain-fed agriculture, and limited agro-processing infrastructure. These difficulties are further exacerbated by climate change, which manifests in unpredictable rainfall patterns and extreme weather events, posing serious threats to crop yields and overall food availability (Koryo-

Dabrah et al., 2021). These challenges often lead to unstable food production and prices, affecting vulnerable populations. Nutritional status, closely linked to food security, is another critical concern. In Ghana, malnutrition remains a concern, particularly among children under five years old. The 2020 Comprehensive Food Security and Vulnerability Analysis (CFSVA) report highlights that stunting, wasting, and underweight conditions are prevalent in specific regions, significantly impacting children's physical growth and overall development (Ghana Statistical Service, 2020). This research seeks to evaluate Ghana's current food security situation and nutritional well-being, pinpointing crucial contributors to food insecurity and malnutrition, while recommending potential solutions to address these challenges.

1.2. Problem Statement

Over the past decade, Ghana has made remarkable strides in enhancing food and nutrition security, which has contributed to the country's progress toward achieving the Millennium Development Goals related to poverty and hunger reduction (Ecker & Asselt, 2017). Despite these advancements, persistent challenges remain, including regional inequalities and the growing dual burden of malnutrition—where both undernutrition and overnutrition, including micronutrient deficiencies, coexist (Ecker & Asselt, 2017). The prevalence of overweight and obesity among adults is increasing rapidly across urban and rural areas, heightening the risk of noncommunicable diseases. Furthermore, food insecurity and its associated health complications continue to affect individuals and households in distinct ways, influenced by social and economic disparities (Asuru, 2020).

Critical questions remain regarding the effectiveness of current food security programs, the impact of market fluctuations on food prices, the role of dietary diversity in improving nutritional outcomes, and the influence of socio-cultural factors on food choices and practices. Therefore, this study seeks to address the critical need for a comprehensive analysis of food security and nutritional status in Ghana, to identify the key challenges, and to provide evidence-based recommendations for more effective policies and interventions that promote equitable access to nutritious food for all Ghanaians.

1.3. Significance of Study

This study is crucial for Ghana because it provides a detailed look at food security and nutrition, informing practice, policy, and academia. For practitioners, it pinpoints specific areas and populations needing targeted support. For policymakers, it offers evidence to create effective food security programs, emphasizing the inclusion of farmers' voices. Academically, it expands the understanding of food security in Ghana, acting as a reference for future research and educational programs in agriculture, nutrition, and development.

2 AIM AND METHODOLOGY

2.1. Aim of the Study

The primary objective of this study is to provide a comprehensive analysis of food security and nutritional conditions in Ghana. Specifically, it aims to:

1. Investigate the current state and trends of food security in Ghana, considering key aspects such as availability, affordability, quality, safety, sustainability, and adaptation.
2. Examine the major factors influencing food security and nutritional well-being, including the identification of critical constraints within the food system.
3. Assess existing government policies, interventions, and progress in addressing food security challenges in Ghana.
4. Evaluate Ghana's advancements toward meeting the United Nations Sustainable Development Goals (SDGs) 1, 2, and 3 by 2030.

2.2. Research Questions to be Answered

This research aims to address the following key questions:

1. What are the past trends and status of food security in Ghana, particularly in relation to availability, affordability, quality, safety, sustainability, and adaptation?
2. What are the primary factors influencing food security and nutritional conditions in Ghana, and what systemic constraints hinder progress?
3. How effective have government policies and interventions been in tackling food security challenges in Ghana?
4. To what extent is Ghana progressing toward achieving the United Nations Sustainable Development Goals (SDGs) 1 (No Poverty), 2 (Zero Hunger), and 3 (Good Health and Well-being) by 2030?

2.3. Research Methods

This chapter outlines the methodology used in this study to assess the current state of food security and nutritional status in Ghana. It describes the research approach, design, data sources, data collection and analysis procedures, and the ethical considerations associated with using secondary data. The chosen methods aim to provide a comprehensive and reliable analysis that can effectively inform policy and intervention strategies.

2.3.1. Sources of Data

The study relies on a broad selection of secondary data sources. National surveys, such as the Ghana Statistical Service's Demographic and Health Surveys (DHS), provided in-depth household-level insights into food security, nutritional status, and socio-economic conditions. Furthermore, reports from global organizations like the Food and Agriculture Organization (FAO) and the World Health Organization (WHO), along with data from Ghana's Ministry of Food and Agriculture (MoFA), offered a broader perspective on food security trends and policy impacts. In addition, academic research from universities and institutions contributed valuable findings and datasets, enriching the depth and accuracy of the study's analysis and offered insights into the broader impacts of food security policies and trends. Academic research from universities and institutions was also reviewed to incorporate findings and datasets that give depth to the study's analysis.

2.3.2. Data Analysis Procedures

The analytical process will involve both descriptive and trend analysis. Descriptive statistics will summarize the data through frequencies, means, and percentages, outlining the extent of food security and various nutritional outcomes across the population. Trend analysis will identify historical changes and patterns in food security and nutrition. These analyses will pinpoint key areas for intervention and policy development.

2.3.3. Limitations of the Study

The study on food security and nutritional status in Ghana faces several limitations:

Data Limitations: Utilizing secondary data may lead to gaps in coverage, particularly from remote areas or less-represented population groups. This limits the scope and accuracy of the findings.

Scope of Study: The retrospective nature of secondary does not capture recent developments affecting food security and nutrition, limiting the study's relevance to current conditions.

Generalization of Findings: Results are not generalizable across all regions or demographic groups in Ghana due to significant regional disparities in food security and nutritional status.

2.4. Structure of the Work

The study is structured into five core chapters. Chapter One provides an overview of food security in Ghana, presenting the contextual background, problem statement, and significance of the research. Chapter Two outlines the study's objectives and guiding research questions, detailing the research design, data collection methods, analytical techniques, and ethical considerations. Chapter Three lays the theoretical foundation of food security, exploring key concepts and frameworks of food security, examines global and regional trends, explores key dimensions of food security in Ghana, reviews government policies, and identifies research gaps the study addresses.

Chapter four analyses Ghana's current food security status across all dimensions, identifies major constraints, assesses government interventions, evaluates progress toward SDGs 1, 2, and 3, and compares findings with neighbouring West African countries. Chapter five summarizes key findings, draws conclusions addressing the research questions, outlines policy implications, provides targeted recommendations for stakeholders, suggests directions for future research, and offers final perspectives on Ghana's food security outlook.

3 LITERATURE REVIEW

The literature review provides a comprehensive examination of food security, exploring its definitions, key dimensions, and its vital role in achieving Sustainable Development Goals (SDGs), particularly SDG 2: Zero Hunger. It analyses core elements of food security—including availability, accessibility, utilization, and stability—while evaluating the various indicators used to measure food security at national, household, and individual levels. The review also addresses the pressing issue of food insecurity, providing insights into its prevalence and underlying causes in different regions, including a detailed focus on Africa and Ghana. Additionally, it highlights the crucial aspect of nutritional status and its impact on health outcomes, emphasizing the prevalence of nutrition-based diseases in Ghana. The review concludes with a discussion of relevant theories including the Entitlement Theory and the Food Systems Approach, offering a deeper understanding of the complexities involved in food security and the strategies needed to tackle food insecurity effectively.

3.1. Review of Relevant Theories

While numerous theories examine food security and development, the Entitlement Theory and Food Systems Approach offer unique and complementary perspectives for understanding complex food-related challenges.

3.1.1 Entitlement Theory

The Entitlement Theory, pioneered by Nobel laureate Amartya Sen, offers a comprehensive framework for analyzing food security beyond traditional production-based approaches. In the context of Ghana, this theory provides critical insights into the complex mechanisms of food access and nutritional status (Sen, 1982). The Entitlement Theory posits that food security extends beyond mere food availability, encompassing a complex interaction of economic, social, and political entitlements that shape an individual's capacity to obtain food.

Fundamentally, the theory examines how individuals and households access food through various mechanisms. These include entitlements based on production (the ability to cultivate food), trade (the ability to exchange goods or services for food), labour (income earned through work), and transfers (assistance received from governmental bodies or social networks).

(Devereux, 2001). In Ghana, where economic vulnerabilities and agricultural challenges persist, this approach provides a nuanced understanding of food insecurity that goes beyond simplistic production metrics.

The theory's significance in Ghana's context lies in its ability to unpack the structural barriers that prevent consistent food access. By examining the intricate pathways through which individuals secure food, researchers can identify specific vulnerabilities within different population groups (Dreze & Sen, 1990). For instance, the theory helps explain why some households might struggle with food security despite national-level agricultural production, highlighting the importance of economic rights and social protection mechanisms.

Methodologically, the Entitlement Theory enables a comprehensive analysis of food security by mapping multiple dimensions of access. It shifts the focus from aggregate food availability to individual and household-level capabilities, providing a more dynamic and contextually sensitive approach to understanding nutritional challenges (Maxwell & Smith, 1992). In Ghana, this means examining not just agricultural output, but also factors like income distribution, labor markets, social protection programs, and economic policies that influence food entitlements.

Empirical applications of the theory in Ghana can reveal critical insights into nutritional disparities. Researchers can identify the most vulnerable populations and design targeted interventions by analyzing how different socioeconomic groups develop strategies to access food. The approach recognizes that food insecurity is not simply a result of food scarcity, but a complex outcome of economic and social inequalities (Sen, 1982).

3.1.2 Food System Approach

In Ghana, the Food Systems Approach offers a comprehensive perspective on nutritional challenges by evaluating the entire food system, from production to consumption (Ericksen, 2008).

Unlike conventional methods that primarily focus on agricultural production, this approach considers various factors that influence food security. These include environmental conditions, economic structures, social dynamics, and policy frameworks that collectively impact food availability, accessibility, and utilization (HLPE, 2017). Applying this framework to Ghana highlights the interplay between climate change, economic policies, agricultural practices, and social inequalities in shaping nutritional outcomes.

Pinstrup-Andersen and Herforth a proponent of the food system approach, argue that food security cannot be understood through isolated interventions (Pinstrup-Andersen, 2021). The approach demands a multidisciplinary perspective integrating agricultural, nutritional, economic, and social analyses. In Ghana, this means examining how agricultural policies, market structures, infrastructure, and cultural practices collectively influence food security.

The methodology involves mapping the entire food system, identifying critical intervention points, and understanding the complex interactions between different components. This includes analyzing food production, processing, distribution, marketing, consumption, and waste management (CGIAR, 2016). For Ghana, such an approach can reveal nuanced insights into why nutritional challenges persist despite agricultural potential.

The Food Systems Approach facilitates more precise and well-rounded interventions. By recognizing the interconnected factors contributing to food insecurity, policymakers can craft holistic solutions that target fundamental issues rather than just their manifestations. In Ghana, this could involve comprehensive strategies that simultaneously enhance agricultural output, improve market accessibility, promote nutrition awareness, and strengthen social protection mechanisms (FAO, 2018).

3.1.3 Malthusianism

Thomas Robert Malthus (1766-1834) developed his influential theory in the 1798 "Essay on the Principle of Population," challenging the optimistic Enlightenment views of thinkers like Godwin and Condorcet (Foster, 2000). His theory established a framework for understanding the relationship between population growth and resource constraints that continues to shape food security discussions. Malthus proposed two fundamental principles: population grows exponentially (1, 2, 4, 8...), while food production increases only arithmetically (1, 2, 3, 4...), creating an inevitable mathematical mismatch (Malthus, 2023). This discrepancy formed the basis of his prediction that population would eventually outstrip food supply, leading to "positive checks" like famine, disease, and war. Malthus wrote, "The power of population is so superior to the power of the earth to produce subsistence for man, that premature death must in some shape or other visit the human race" (Malthus, 2023, p. 44).

He identified two categories of population checks: "preventive checks" (voluntary choices like delayed marriage) and "positive checks" (involuntary forces like famine and disease) that

maintain balance between population and resources (Winch, 2013). This framework established a way of understanding demographic dynamics that influenced scholarly thought for generations. Malthusian theory draws strength from historical observations. Before the Industrial Revolution, population growth was indeed often constrained by food supply (Livi-Bacci, 2017). The theory acknowledges ecological limits in a manner that resonates with modern sustainability science. As Ehrlich and Ehrlich (2009) note, Malthus recognized fundamental ecological principles regarding carrying capacity long before formal ecological science emerged.

However, Malthusian theory has significant weaknesses. Most notably, Malthus underestimated humanity's capacity for technological innovation. The Green Revolution and other agricultural advances dramatically increased yields beyond what he considered possible (Pingali, 2012). Boserup (1965) argued that population pressure itself stimulates innovation, creating a feedback mechanism Malthus overlooked. The theory also failed to anticipate the demographic transition, whereby prosperity and education lead to voluntary fertility reduction (Kirk, 1996). Another critical weakness is Malthus's neglect of distribution issues. As Sen, (1982) demonstrated, food insecurity often results from inequitable distribution rather than absolute scarcity, with famines occurring amid sufficient aggregate food supplies. The deterministic nature of Malthus's model also failed to account for human adaptability and social innovation that have repeatedly defied predictions of demographic catastrophe (Lee, 2002).

Malthusianism established a scarcity paradigm that framed food security primarily as a supply problem, influencing policy approaches throughout the 20th century (Lang & Barling, 2012). This led to population control advocacy in development programs and the productivity-focused Green Revolution (Evenson & Gollin, 2003). Malthus's concept of carrying capacity became a cornerstone of ecological thinking about food production, proving valuable in regional contexts even as it was rejected as a global determinant of food security (Daily & Ehrlich, 1992). While pure Malthusianism has been largely discredited globally, elements remain relevant to modern challenges. Some regions still face genuine supply constraints, particularly areas with rapid population growth and limited arable land (Clover, 2003). Climate change, water scarcity, and soil degradation create productivity challenges that revive interest in Malthusian constraints (Godfray et al., 2010). As Brown (2011) suggests, emerging ecological limits may force reconsideration of Malthus's core insights despite technological advances.

Contemporary approaches to food security have evolved beyond Malthusian foundations. The food sovereignty movement emphasizes people's right to define their own food systems (Patel, 2009). Sustainable intensification aims to increase production while reducing environmental impacts (Pretty, 2010). Modern frameworks emphasize nutritional quality rather than just caloric sufficiency and focus on building resilient systems rather than maximizing short-term production (Tendall et al., 2015). Despite its limitations, Malthus's theory continues to influence policy and provoke debate. According to Trewavas (2002), its enduring value may lie not in predictive accuracy but as a cautionary perspective that encourages critical examination of assumptions about unlimited growth. Modern approaches acknowledge resource constraints while recognizing that technology, social organization, and policy choices significantly mediate the relationship between population and food supply (Lang et al., 2009).

3.2. The Concept of Nutritional Status

Nutritional status refers to the condition of an individual's health as influenced by the intake and utilization of nutrients (Sauberlich, 2018). It is a crucial aspect of overall health and well-being, affecting physical growth, cognitive development, and the ability to fight infections and diseases (Gibson, 2005). Nutritional status encompasses various dimensions, including dietary intake, nutrient absorption, as well as the human metabolism system.

3.2.1. Dietary Intake

Dietary intake plays a crucial role in determining nutritional status. It involves consuming various foods and beverages that supply essential nutrients, including carbohydrates, proteins, fats, vitamins, and minerals. A balanced intake supports optimal bodily functions, while deficiencies or excesses can lead to malnutrition. Malnutrition manifests in two forms: undernutrition, which includes stunting, wasting, and micronutrient deficiencies, and overnutrition, which contributes to obesity and related non-communicable diseases (WHO, 2020).

3.2.2. Nutrient Absorption

The careful process whereby the body utilizes the nutrients from the food we eat is termed nutrient absorption. This process occurs in the small intestine, where nutrients are absorbed into the bloodstream and transported to various cells and tissues. Factors such as

gastrointestinal health, enzyme activity, and the presence of certain medical conditions can affect nutrient absorption. Poor nutrient absorption can lead to deficiencies even if dietary intake is adequate (Gibson, 2005).

3.2.3. Physiological and Metabolic Functions

The body's physiological and metabolic functions play a significant role in determining nutritional status. These functions include digestion, metabolism, and regulating nutrient levels in the body. Metabolic disorders, chronic diseases, and infections can impact these functions and affect nutritional status. For example, conditions such as diabetes, anemia, and intestinal infections can disrupt the normal absorption and utilization of nutrients (Haddad et al., 2015).

Assessing nutritional status involves evaluating dietary intake, anthropometric measurements, biochemical indicators, and clinical signs. Anthropometric measurements, such as height, weight, and body mass index (BMI), provide insights into physical growth and body composition. Biochemical indicators, including blood and urine tests, help identify nutrient deficiencies or excesses. Clinical signs, such as changes in skin, hair, and nails, can also indicate nutritional imbalances (WHO, 2020).

3.3. The Concept of Food Security

Food security is a complex issue that has drawn considerable attention from researchers, policymakers, and field experts. The World Food Summit (1996) defines it as a condition where everyone always has reliable physical and economic access to adequate, safe, and nutritious food that aligns with their dietary needs and preferences for a healthy and active life. This definition highlights not only food availability but also accessibility and quality as essential factors.

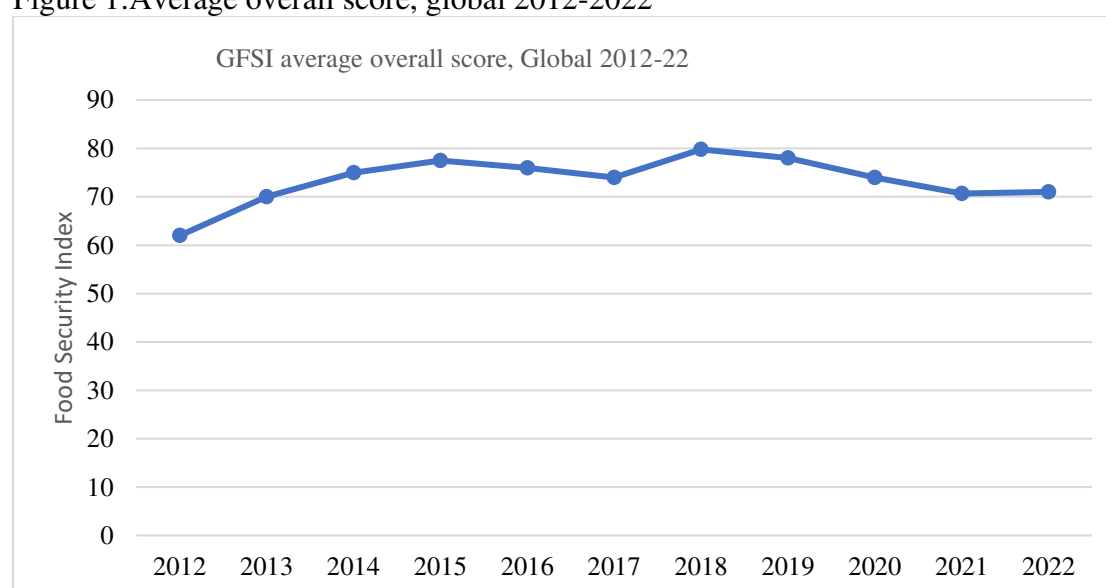
Expanding on this concept, the World Bank (1986) describes food security as ensuring that all individuals have consistent access to sufficient food to sustain an active, healthy life. Additionally, the World Bank differentiates between chronic and transitory food insecurity—chronic food insecurity being a long-term challenge due to an ongoing inability to secure adequate food, whereas transitory food insecurity arises from temporary disruptions such as fluctuations in food prices, production, or household income, thereby affecting access to sufficient food.

Household food security extends beyond food access to vital components such as a balanced diet, safe drinking water, environmental sanitation, primary education, and essential healthcare for all members of a household.

3.4. Dimensions of Food Security

Food security has four main dimensions: availability, access, utilization, and stability. At a time when ensuring food security is critical globally, the Global Food Security Index (GFSI) indicates a worsening global food landscape (Figure 1). After reaching its highest point in 2019, the GFSI has been on a downward trend due to surging food prices and a sharp rise in hunger levels. Spanning over a decade of data, the index reveals a gradual weakening of the global food system, with major disruptions between 2020 and 2022 such as the COVID-19 pandemic and soaring commodity prices exposing its vulnerabilities. These events have amplified the deep-rooted challenges threatening food security and undermining the system's ability to withstand future shocks.

Figure 1: Average overall score, global 2012-2022



Source: Global Food Security Index, 2022

After climbing year on year between 2012 to 2018, the overall food security score has not improved since 2019

The decline marks a reversal from the index's initial years, which saw eight years of consistent improvement before progress began to slow. This stagnation highlights underlying structural challenges and growing risks, including fluctuations in agricultural output, depletion of natural

resources, widening economic inequality, and instability in trade and supply chains. Recent economic and political disruptions have further strained an already fragile system. As such crises grow more intense and frequent, the threat to global food security continues to rise.

3.4.1. Availability

Availability refers to the supply of food in each area. Agricultural production, food imports, and food aid influence it. The availability of food is essential for ensuring that there is enough food to meet the population's needs. Studies have shown that food availability is affected by climate change, natural disasters, and economic conditions (Ashby, Kleve, McKechnie, & Palermo, 2016). According to the Food and Agriculture Organization of the United Nations (2024), the global per capita dietary energy supply reached approximately 2,985 kilocalories per person per day in 2022, reflecting a 5% increase since 2010. This improvement suggests that, on average, more food is available for human consumption worldwide than a decade ago. However, this global average masks significant disparities between regions and countries. While some areas have seen substantial gains in food supply, others, particularly those affected by conflict, climate shocks, or economic instability, continue to face persistent shortages.

The Economist Intelligence Unit's Global Food Security Index ,(GFSI) further contextualizes these figures by assessing food availability across 113 countries. The table below shows the ranking and score of selected countries rank and their score on food availability out of total score of 100.

Table 1: Selected countries' rank and score on food availability 2022

Country	Position on a list of 133 countries	score of food availability on a scale of 0 to 100
Finland	15 th	70.5
Sweden	21 st	68.3
Czech Republic	17 th	69.4
South Korea	11 th	71.5
Malaysia	56 th	59.5
Yemen	112 th	26.9
South Africa	52 nd	60.1
Nigeria	108 th	39.5
Ghana	80 th	52.4
Sierra Leone	110 th	35.5

Source: The Economist Intelligence Unit.

Political instability and armed conflict threats have increased over the last three years, according to the GFSI, which has seen scores decline by 6% and 4%, respectively. Alongside this, there has been an increase in reliance on chronic food assistance, whose score has decreased by 8% since 2019 (The Economist Group, 2022). These shocks have exposed vulnerabilities in food systems, particularly in regions with limited access to imports or agricultural technology. High-income countries like the Finland and South Korea consistently achieve the high GFSI scores, reflecting strong food availability, while many developing countries like Yemen and Sierra Leone remain at risk due to lower agricultural productivity and greater reliance on food imports

3.4.2. Access

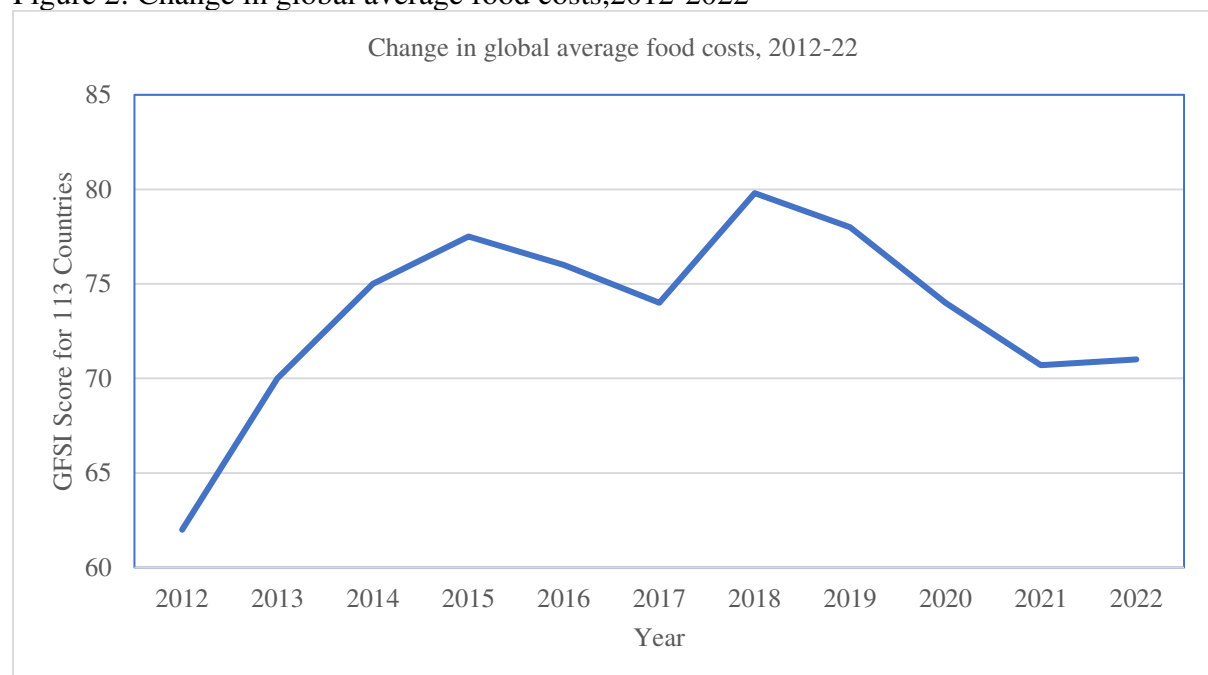
Access includes the ability of individuals and households to obtain food. This dimension includes economic access (the ability to afford food); physical access (the proximity to food sources); and social access (the ability to secure food without discrimination). Access to food is driven by income levels, food prices and transportation infrastructure (Ashby et al.,2016).

According to the State of Food Security and Nutrition in the World (SOFI) report by FAO, in 2023, about 2.333 which is approximately 28.9% of the world's population faced moderate to severe food insecurity, meaning limited access to foods.

Of these, 864 million people experienced severe food insecurity, often going without food for an entire day or more. This level of food insecurity has remained persistently high since 2020, reflecting barriers to food access despite global efforts (FAO, IFAD, UNICEF, 2023).

Income levels and food prices heavily influence economic access to food. The USDA's International Food Security Assessment (IFSA) model estimated that in 2023, nearly 229 million fewer people were food insecure compared to 2022, a 16.8% decrease, largely due to average income growth of 3.7% in low- and middle-income countries and a drop in vegetable oil prices, which are a significant calorie source in many diets. However, this improvement is uneven, and many regions continue to struggle with affordability and access (Cardell & Zereyesus, 2023). Affordability lowers the GFSI 2022 index. Between 2019 and 2022, the index's affordability score decreased by 4%, from 71.9 to 69, as food prices increased as a result of shocks like the COVID-19 epidemic and the war in Ukraine. Additionally, people all around the world are finding it more difficult to acquire food due to dwindling trade freedom and a lack of funding for safety nets.

Figure 2: Change in global average food costs,2012-2022



Source: Global Food Security Index,2022

Food prices have skyrocketed between 2019 and 2022, as evidenced by the dismal GFSI ratings that measure average food costs performance has fallen by 11.4% as shown in Figure 2. Policymakers are anxious to prevent a recurrence of the 2008 food price crisis, which also saw

record highs, as the world currently faces its third global food price crisis in 15 years (Food and Agriculture Organization of the United Nations, 2022). However, the challenge they face is formidable. According to the International Panel of Experts on Sustainable Food Systems, "conflicts, poverty, and climate change are now combining to create endemic and widespread' risks to global food security," meaning that unless steps are taken to curb the threats, higher food prices may become the new normal.

Physical access to food is affected by infrastructure and proximity to food sources. Urbanization is reshaping food systems globally, with projections that by 2050, 70% of the world's population will live in cities. This shift increases reliance on markets and supply chains, which can be disrupted by conflicts, climate shocks, or economic crises, thereby limiting physical access to food for vulnerable populations. Rural areas, especially in Africa and Asia, are increasingly dependent on national and global food markets, making them susceptible to price volatility and supply disruptions (FAO, IFAD, UNICEF, 2023). Social access, including the ability to obtain food without discrimination, also plays a role but is less quantified in global data. However, marginalized groups, including women and rural residents, are disproportionately affected by limited access to nutritious and sufficient food.

Table 2:Selected countries' rank and score on food access 2022

Country	Position on a list of 113 countries	Percentage of Food access on a scale of 0 to 100
Finland	7 th	91.9
Sweden	7 th	91.9
Czech Republic	11 th	91.3
South Korea	51 st	76.8
Malaysia	30 th	87.0
Yemen	95 th	46.4
South Africa	70 th	63.4
Nigeria	113 th	25.0
Ghana	75 th	59.9
Sierra Leone	104 th	36.6

Source: The Economist Intelligence Unit.

Regionally, food insecurity is most severe in Africa as shown in the table for Nigeria and Sierra Leone, where 58% of the population faces moderate or severe food insecurity, nearly double the global average. Asia, Latin America, and Oceania have lower but still significant prevalence rates of 24.8%, 28.2%, and 26.8%, respectively (UNICEF, 2021). High income countries like Finland, Sweden and Czech Republic have high percentage score on food access due to economic structures present in these advanced countries. Acute food crises also highlight extreme access challenges. In 2023, nearly 282 million people across 59 countries faced high levels of acute food insecurity requiring urgent assistance. The Gaza Strip, for example, experienced the most severe food crisis recorded, with the entire population facing crisis conditions or worse, and a quarter facing catastrophic levels of food insecurity (Food Security Information Network) and Global Network Against Food Crises., 2023).

3.4.3. Utilization

Utilization refers to the body's ability to absorb and use nutrients from food. It is affected by factors such as food quality, dietary diversity, and health status. Proper utilization of food is essential for maintaining good health and preventing malnutrition. Studies have highlighted the importance of nutrition education and access to healthcare in improving food utilization (Manikas, Ali & Sundarakani, 2023). According to the GFSI report by The Economist Group (2022), the quality and safety pillar's scores dropped from 67.1 to 65.9 as a result of the significant declines in nutritional standards, especially in national nutrition programs and monitoring. In 2022, nations from across the world failed to meet their nutritional goals. In 2022, around one-third of nations (35 out of 113) did not have a national nutrition plan or strategy, which is almost twice as many as in 2019. Furthermore, out of 113 nations, 25 do not routinely assess the nutritional status of their citizens (up from 15 in 2019). Policymakers are unable to detect dietary inadequacies and allocate resources, when necessary, with routine monitoring.

The table below shows the ranking and scores of selected countries when it comes to food utilization.

Table 3: Selected countries' rank and score on food utilization 2022

Country	Position on a list of 113 countries	Percentage of Food Utilization on a scale of 0 to 100
Finland	4 th	88.4
Sweden	11 th	85.0
Czech Republic	32 nd	76.3
South Korea	50 th	71.5
Malaysia	38 th	74.7
Yemen	98 th	48.7
South Africa	60 th	66.1
Nigeria	79 th	55.6
Ghana	94 th	50.5
Sierra Leone	109 th	41.8

Source: The Economist Intelligence Unit.

Out of 113 countries, Ghana ranked 94th in food utilisation. Looking at the country's rank at 80 in food availability, it is not doing enough to fully utilise the food available to meet the nutritional needs of the citizens. Though Finland's food availability score was 70.5, it is crucial to note that its utilisation score is 88.4. Technological innovation and programs on food processing, storage, and distribution are essential in combating food insecurity.

3.4.4 Stability

Stability relates to the consistency of food supply over time. It is affected by factors such as political stability, economic stability, and environmental conditions. A stable food supply is crucial for ensuring that individuals and households can access food consistently, without facing disruptions due to external shocks (Ashby et al., 2016). Countries must prioritize agricultural adaptation strategies to achieve long-term, sustainable solutions, rather than just reacting to shocks. The 2022 GFSI shows a 3.8% increase in the sustainability and adaptation pillar, driven by increased political commitments, national agricultural adaptation policies, disaster risk management, and environmental economic accounting. Overall scores have plateaued since 2019.

Table 4: Selected countries' rank and score on food stability 2022

Country	Position on a list of 113 countries	Percentage of Food Stability on a scale of 0 to 100
Finland	2 nd	82.6
Sweden	14 th	68.3
Czech Republic	8 th	70.3
South Korea	34 th	58.5
Malaysia	57 th	53.7
Yemen	105 th	37.8
South Africa	44 th	56.9
Nigeria	57 th	53.7
Ghana	88 th	45.1
Sierra Leone	73 rd	49.8

Source: The Economist Intelligence Unit.

Food stability depends on how the natural resources are managed and sustained to enable them to provide for the needs of the present and future generations. Ghana's score in food stability is weakest in the four food dimensions. This is a projection into the future of the country's food security decline. Finland has adequate measures in place to handle disaster shocks without collapsing its food production system and has good natural resource protection (protection of land against erosion, deforestation, bush fires, as well as ocean, river and lake health) as compared to low-ranked countries like Yemen, Ghana and Sierra Leone.

For food security to be fully realized, all four dimensions must be met simultaneously. Additionally, nutrition is integral to food security, which is achieved when secure food access is combined with a sanitary environment, adequate health services, and informed care practices.

3.5. Levels of Food Security Measurement

Food security can be evaluated at three key levels: national, household, and individual, each offering distinct insights into the broader food security landscape and highlighting specific areas that require attention.

At the national level, food security is primarily assessed through supply and demand indicators. These include factors such as agricultural productivity, food production volume, stock levels, and trade balances in food imports and exports. Food production reflects the total amount of food grown domestically, while imports and exports indicate the extent to which international trade influences overall food availability (Porkka et al., 2013). Food stocks indicate the amount stored for future use, which helps buffer against supply shocks. Agricultural productivity, measured as yield per hectare, assesses the efficiency of food production processes (Food and Agriculture Organization., 2014). Supply and demand indicators assess food availability at a macro level, providing a comprehensive picture of the country's food security situation. They help policymakers make informed decisions regarding food policies and resource allocation.

Food security can be measured at the household level through direct dietary intake surveys. These studies assess the type and quantity of food households consume and provide insight into the diversity and suitability of their diet. The most important indicators at this level include the Household Dietary Diversity Score (HDDS), Food Consumption Score (FCS), and the Household Food Insecurity Access Scale (HFIAS). HDDS measures a variety of foods consumed in the home over a period of time, while FCS assesses the frequency and nutritional quality of foods consumed (Leroy et al., 2015). HFIAS measures the prevalence and severity of food insecurity based on budget experience and coping strategies (Humanitarian Global, 2021). These indicators help identify households at risk of food insecurity and malnutrition. Therefore, targeted interventions can improve nutritional and food security.

Food security at the individual level can be measured by anthropometric information, including measurements of the physical dimensions of the body to assess nutritional status (Lele et al., 2016). The most important indicators at this level include Height- for-Age (stunting), Weight-for-Height (waste), and Body Mass Index (BMI). Height- for-Age measures chronic malnutrition by comparing the child's size with an age standard, while the Weight-for-Height assesses acute malnutrition by comparing the child's weight to height standards. BMI measures nutritional status in adults by comparing weight to height (UN, 2016). Anthropometric indicators provide valuable information on the nutritional status of vulnerable or high-risk people, especially children and pregnant women. They help identify people who need nutritional support and interventions.

The per capita supply of food grains is an indicator of food security at both the national and household levels (Singh et al., 2017). It measures the average amount of food grains per person and measures insight into the overall availability of food and consumption patterns. The most important aspects of this indicator include overall food production, population size and per capita availability. The total production of food grains refers to the total amount of food grains produced in a country, and the population size represents the total number of people in the country or households. Per capita availability is calculated by dividing the overall production of food grains by the population size, providing an average amount of food grains per person (Food and Agriculture Organizations, 2014). Per capita supply indicators can help to determine whether the nutritional needs of the population are met through food production and identify potential gaps in food availability.

3.6. Food Insecurity

Food insecurity is defined as inadequate or uncertain economic and social circumstances that affect access to sufficient nutritional foods which promote healthy living (Sassi et al., 2018). Food insecurity remains an important global challenge, affecting millions of people around the world. According to the latest report, the number of people affected around the world increased to 828 million in 2021. Around 2.4 billion people (or 29.6% of the population) worldwide are faced with moderate or severe food uncertainty (Murtoff, 2025). This situation is exacerbated by factors such as conflict, climate variability, and economic slowdowns and fluctuations (Food and Agriculture Organization [FAO], 2022).

The World Bank (2022) Highlights that domestic food price inflation remains high in many low- and middle-income countries. For instance, food price inflation exceeded overall inflation in 43.5% of the 161 countries where data is available. Additionally, the United Nations Office for the Coordination of Humanitarian Affairs (UNOCHA) outlines that more than 280 million people face acute hunger daily (UNOCHA, 2022).

Efforts to address food insecurity are ongoing, with organizations like the FAO, International Fund for Agricultural Development (IFAD), UNICEF, World Food Programme (WFP), and World Health Organization (WHO) working together to monitor and respond to the crisis. These organizations emphasize the importance of sustainable agricultural practices, investments in infrastructure and technology, and social protection programs to reduce poverty and improve access to food (FAO et al., 2022). Despite these efforts, achieving food security

remains a complex and multifaceted challenge that requires coordinated global action and adequate financing. The goal is to ensure that all individuals have access to sufficient, safe, and nutritious food, which is essential for their health and well-being (World Bank, 2022).

3.7. Food Insecurity in Africa

Food insecurity in Africa is a pressing issue affecting millions across the continent. The Food and Agriculture Organization (FAO) estimates that around 346 million people in Africa are impacted by the ongoing food crisis. This issue stems from a combination of challenges, such as economic downturns, political instability, conflicts, and environmental changes. Climate change has had profound effects on food production, with severe weather conditions—including floods and droughts—hindering agricultural activities and causing widespread crop failures. Notable example occurred in 2023 when devastating floods in Nigeria destroyed 850,000 metric tons of food, enough to feed eight million people for six months. Rising global grain prices have also made it difficult for families to afford food (FAO, 2022).

Conflict is another major cause of food insecurity in Africa. Many countries in the region are experiencing ongoing conflicts that have disrupted food supply chains and displaced millions of people. For instance, South Sudan, Ethiopia, and the Central African Republic are among the countries facing severe food insecurity due to conflict (United Nations Office for the Coordination of Humanitarian Affairs [UNOCHA], 2022). Economic shocks, such as the COVID-19 pandemic, have also aggravated African food insecurity. The pandemic has pushed millions of people into extreme poverty, reducing their ability to access food. Additionally, the cost-of-living crisis triggered by Russia's invasion of Ukraine has further strained food supplies and increased prices (World Bank, 2022).

Efforts to address food insecurity in Africa include promoting sustainable agricultural practices, investing in infrastructure and technology, and implementing social protection programs. Organizations like the FAO, World Food Programme (WFP), and International Committee of the Red Cross (ICRC) are working to provide immediate and long-term assistance to those affected by the crisis (FAO et al., 2022). Ensuring food security in Africa remains a formidable challenge, demanding comprehensive strategies, international cooperation, and sufficient financial support. The ultimate aim is to guarantee that every person can obtain an adequate supply of nutritious, safe, and sufficient food—an essential foundation for overall health and well-being (FAO, 2022).

3.8. Food Security and SDG

Food security plays a crucial role in achieving the Sustainable Development Goals (SDGs), particularly SDG 2: Zero Hunger. This goal seeks to eliminate hunger, enhance food security, improve nutrition, and support sustainable agricultural practices by 2030 (United Nations, 2015). To effectively reach this objective, it is essential to recognize the connections between food security and other SDGs, as well as the associated challenges and opportunities.

Food security is deeply intertwined with multiple other goals, including SDG 1 (No Poverty), SDG 3 (Good Health and Well-being), SDG 6 (Clean Water and Sanitation), SDG 13 (Climate Action), and SDG 15 (Life on Land). Addressing food security holistically can create positive ripple effects across these interconnected areas, fostering sustainable development worldwide. For example, poverty reduction (SDG 1) is essential for improving food security, as poverty is a major driver of food insecurity. Similarly, access to clean water and sanitation (SDG 6) is crucial for maintaining food safety and preventing foodborne illnesses. Climate action (SDG 13) is also vital, as climate change poses significant threats to food production and food security (Sporchia et al., 2024).

Achieving food security in the context of the SDGs presents several challenges, including climate change, population growth, and political instability. Climate change significantly impacts food production by triggering extreme weather conditions, altering rainfall patterns, and increasing temperatures (Sekaran et al., 2021). These disruptions can reduce agricultural yields, cause crop failures, and threaten food security, making adaptation and mitigation efforts crucial for sustaining global food supplies. (Sekaran et al., 2021). Increase in population increases the demand for food, putting pressure on natural resources and agricultural systems (Westhoek et al., 2016). Political instability can disrupt food supply chains and heighten food insecurity. However, there are also opportunities to address these challenges through innovative approaches and policies. Sustainable agricultural practices, for example, may reduce the impact of climate change and contribute to improving food production. Investing in infrastructure and technology can improve food distribution and reduce food loss and waste. Guidelines that promote social protection and economic inclusion can help reduce poverty and improve access to food (Wahbeh et al., 2022). Political decisions to achieve SDG 2 and improve nutrition security should pursue a holistic and integrated approach that takes into account the relationship between nutritional safety and other SDGs.

This includes promoting sustainable agricultural practices to improve food production and knocking resilience. Furthermore, social protection programs are required to reduce poverty and improve access to food. They strengthen the conditions of governance and political frameworks to ensure that nutritional safety and nutrition for all is met (Food and Agricultural Organization, 2014). By adopting a holistic, integrated approach, political decisions can work to end hunger and ensure nutritional safety for all

3.9. Food Insecurity in Ghana

Food insecurity in Ghana is an urgent matter, with nearly half of the population experiencing it. According to the 2022 Annual Household Income and Expenditure Survey, 49.1% of the Ghanaian population, approximately 15.1 million people, faced food insecurity in the first quarter of the year. Several factors, including economic challenges, climate change, and inefficiencies in the agricultural sector, drive this situation (Ghana Statistical Services (GSS), 2023)

According to studies, food insecurity concerns in Ghana are particularly acute in rural areas where livelihoods rely heavily on agriculture (Abubakari & Abubakari, 2015). Regarding regional distribution, the Upper East region has the highest level of food insecurity, with up to 28% of its population facing food security issues (Nkegbe et al., 2017). A research conducted by World Food Program in the then-three regions of Northern Ghana discovered that the Upper East region was the most food insecure (Hjelm & Dasori, 2012). In terms of district distribution of food insecurity, Hjelm and Dasori (2012) discovered that the region was home to three of the five districts with the highest rates of food insecurity in Northern Ghana. Other studies have validated the region's reputation as a hotspot of food insecurity (Asare-Nuamah, 2021; Nkegbe et al., 2017). This is a concerning tale that necessitates immediate action to address the country's food security crisis, especially in rural areas.

Efforts to address food insecurity in Ghana include promoting sustainable agricultural practices, improving infrastructure and technology, and implementing social protection programs. The National Food Buffer Stock Company (NAFCO) plays a crucial role in ensuring food availability and price stability by purchasing surplus food from farmers and distributing it to state institutions. The World Food Programme (WFP) also works with the Ghanaian government and other partners to provide food assistance and support to those in need (NAFCO, 2022; WFP, 2022). Despite these efforts, achieving food security in Ghana remains

a complex and multifaceted challenge that requires coordinated action and adequate financing. The goal is to ensure that all individuals have access to sufficient, safe, and nutritious food, which is essential for their health and well-being (FAO, 2022).

3.10. Nutrition Based Diseases and its Prevalence in Ghana

In Ghana, risk factors associated with food-related health outcomes include anemia, micronutrient deficiency, and non-communicable diseases (NCDs), such as hypertension, diabetes, and obesity (Coomson & Aryeetey, 2022). Anemia is a significant public health concern, particularly among children under five years old. A 2014 survey estimated that 66% of young Ghanaian children were anemic (Coomson & Aryeetey, 2022). In 2017, approximately 35.6% of children had anemia, 21.5% had iron deficiency, and 12.2% had iron deficiency anemia (Wegmüller et al., 2020). Anemia and micronutrient deficiencies are prevalent among adolescent girls and women of reproductive age. The 2014 DHS reported 48% anemia among female adolescents aged 15-19 years. In 2017, 21.7% of non-pregnant women were anemic, 13.7% were iron deficient, and 8.9% had iron deficiency anemia (Petry et al., 2019). Anemia and micronutrient deficiencies are more prevalent in rural areas and the Northern Belt of Ghana (Wegmüller et al., 2020).

NCDs, including hypertension, diabetes, and obesity, are significant health concerns in Ghana (Ofori-Asenso et al., 2016). In 2016, NCDs accounted for 37% of all deaths and 43% of deaths in 2010 (World Health Organization (WHO), 2018). Hypertension is among the top 10 causes of admission to hospitals and a leading cause of mortality among adults in Ghana (Nuamah et al., 2017). A 2014 DHS reported a 13% prevalence of hypertension among women and men aged 15-49 years. Overweight and obesity rates are high among women of reproductive age and adolescents, and they are still increasing. Approximately 19.3% of adult women and 5.6% of adult men are living with obesity (Marques et al., 2018). Excessive weight gain and obesity are more common in wealthier, older, and urban women. Diet and alcohol are important risk factors for NCDs in Ghana. Stunting affected 21.4% of children, while wasting affected 7.0% (Agbozo et al., 2016).

RESULTS AND DISCUSSION

4.1. Country Profile

Ghana, a West African nation with a population of approximately 31.7 million people, presents a complex landscape regarding food security and nutrition despite its economic progress over recent decades (Ghana Statistical Service, 2021). The country covers an area of 238,533 km² with diverse ecological zones ranging from coastal savannah to northern Guinea savannah, influencing agricultural production patterns and food availability across different regions (MoFA, 2022). Ghana has experienced significant urbanization in recent years, with approximately 56.7% of the population now residing in urban areas, creating new challenges for food distribution systems and changing dietary patterns (Ghana Statistical Service, 2021).

The Ghanaian economy has demonstrated resilience over the past decade, with agriculture contributing about 18.5% to GDP and employing roughly 29.8% of the workforce (Ghana Statistical Service, 2022). Though Ghana achieved lower-middle-income status in 2011, economic growth has been uneven across regions, influencing disparities in food security outcomes (World Bank, 2021). According to a research by Tuholske et al. (2020), economic growth has not necessarily translated to improved nutritional outcomes, particularly in rural areas where poverty rates remain higher than urban centers.

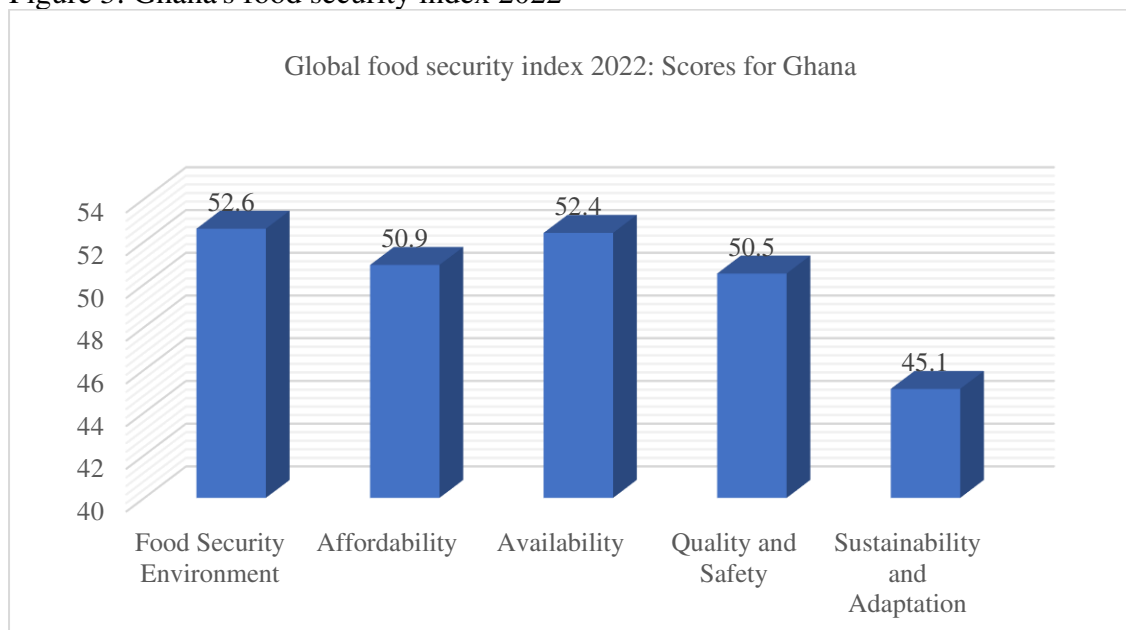
Ghana has made notable progress in food availability at the national level but continues to face persistent challenges in food access, utilization, and stability. The Global Food Security Index ranked Ghana 82nd out of 113 countries in 2022, indicating a moderate food security status with room for improvement (The Economist Impact, 2022). Food insecurity affects approximately 12.8% of households at the severe level and 35.4% at the moderate level, with significant regional disparities that reflect historical development patterns and ecological conditions (WFP & MoFA, 2020). The northern regions experience substantially higher rates of food insecurity (46.7%) compared to southern regions (13.8%), demonstrating geographical inequalities in food access and availability (Nikoi & Antwibosiakoh, 2019).

4.2. The Current Trends and Status of Food Security in Ghana

The analysis of Ghana's food security based on the Global Food Security Index (GFSI) 2022 reveals a complex landscape of modest achievements and significant challenges. Figure 3

provide a clearer picture of Ghana's performance across the various dimensions of food security.

Figure 3: Ghana's food security index 2022

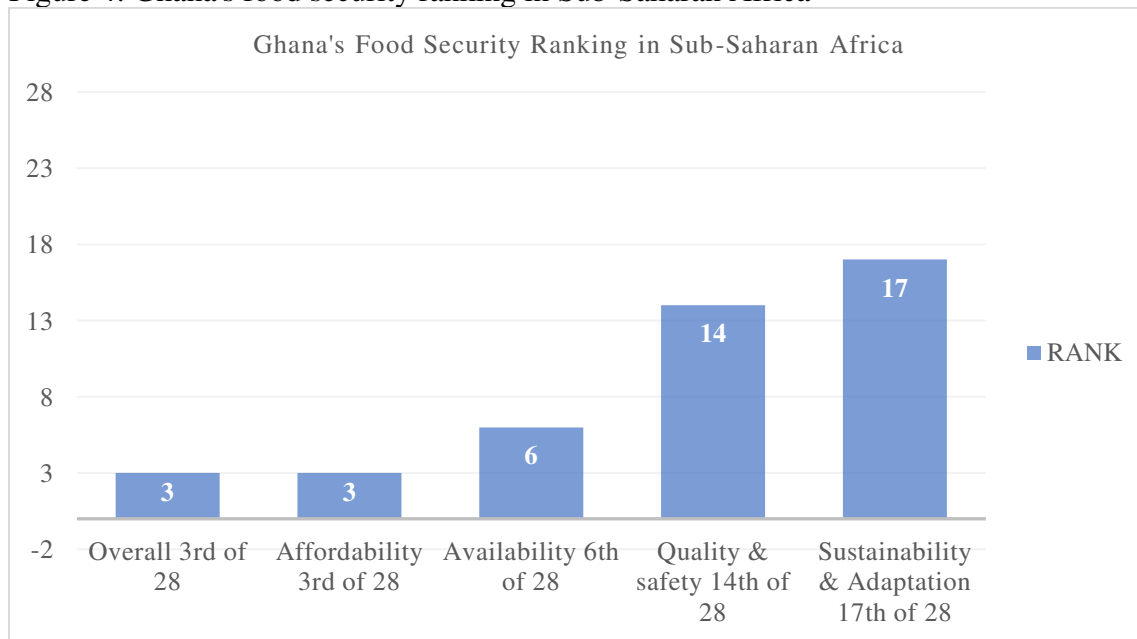


Source: Global Food Security Index, 2022

4.2.1 Food Security Environment

Ghana's overall food security environment presents a mixed picture, with some progress over the past decade but significant challenges remaining. According to the 2022 Global Food Security Index (GFSI), Ghana ranks 83rd out of 113 countries globally, with an overall food security score of 52.6 out of 100. While this places Ghana among the top performers in the Sub-Saharan African region (3rd out of 28 countries) (Figure 4), it still falls below the global average.

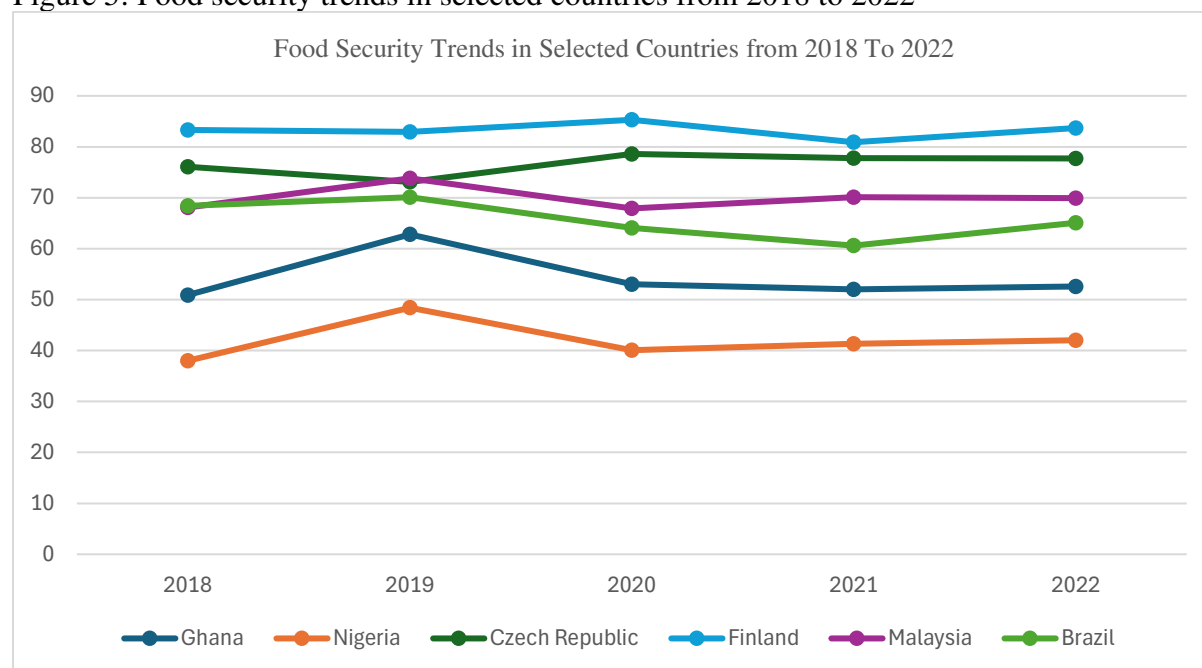
Figure 4: Ghana's food security ranking in Sub-Saharan Africa



Source: Global Food Security Index, 2022

Though it seems Ghana is doing well in Sub-Saharan Africa, in comparison with other countries as shown in Figure 5, from well advanced economies, the country is doing poorly when it comes to food security. After the COVID 19 pandemic which affected most countries, Ghana has not been able to pick up at a promising pace.

Figure 5: Food security trends in selected countries from 2018 to 2022



Source: Global Food Security Index, 2022

This slow progress indicates that while Ghana has made some advancements in strengthening its food security systems, the pace of improvement has been significantly low almost unnoticeable in the scores.

Ghana's food security landscape must be analysed within the broader regional context of Sub-Saharan Africa, which scores an average of 47 points on its overall food security environment in the 2022 GFSI, making it the least food-secure region globally. Ghana's score of 52.6, while higher than the regional average, still represents a "weak" performance according to the GFSI scoring brackets.

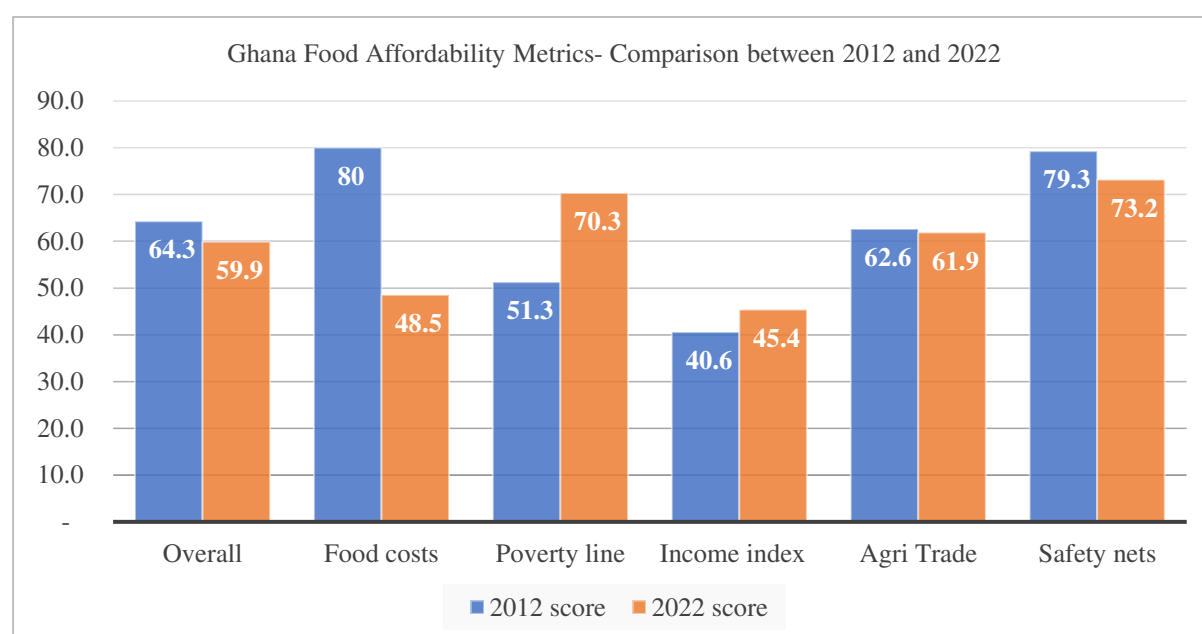
The relatively stagnant trajectory of Ghana's food security environment over the past decade raises important questions about the effectiveness of existing policies and interventions. While there have been improvements in specific areas, particularly in farm infrastructure and food safety, these gains have been partially offset by deterioration in other dimensions, such as food affordability and nutritional standards.

4.2.2 Affordability

Affordability represents Ghana's strongest pillar in the food security index, with a score of 59.9 out of 100. This places the country 75th globally and 3rd regionally. Despite being Ghana's

best-performing component, the Affordability pillar has actually declined by 4.4 points since 2012, indicating a concerning trend in food affordability for Ghanaian consumers. As shown in figure 6, the deterioration in affordability can be primarily attributed to increases in average food costs in the country. The indicator for "Change in average food costs" has declined dramatically by 31.5 points since 2012, reflecting significant food price inflation over the decade. This trend presents a serious challenge to food security, as rising food costs can push vulnerable populations into food insecurity.

Figure 6: Ghana's food affordability metrics (2012 vs 2022)



Source: Global Food Security Index, 2022

Despite these challenges, there are some positive aspects within the Affordability pillar. Ghana scores in the "good" range (70-79.9) for two critical indicators: "Proportion of population under global poverty line" (70.3) and "Food safety-net programmes" (73.2). The relatively strong performance in these indicators suggests that Ghana has made progress in reducing extreme poverty and implementing social protection mechanisms that buffer vulnerable populations against food insecurity.

The improvement of 19 points in the "Proportion of population under global poverty line" indicator since 2012 is particularly noteworthy, indicating substantial progress in poverty

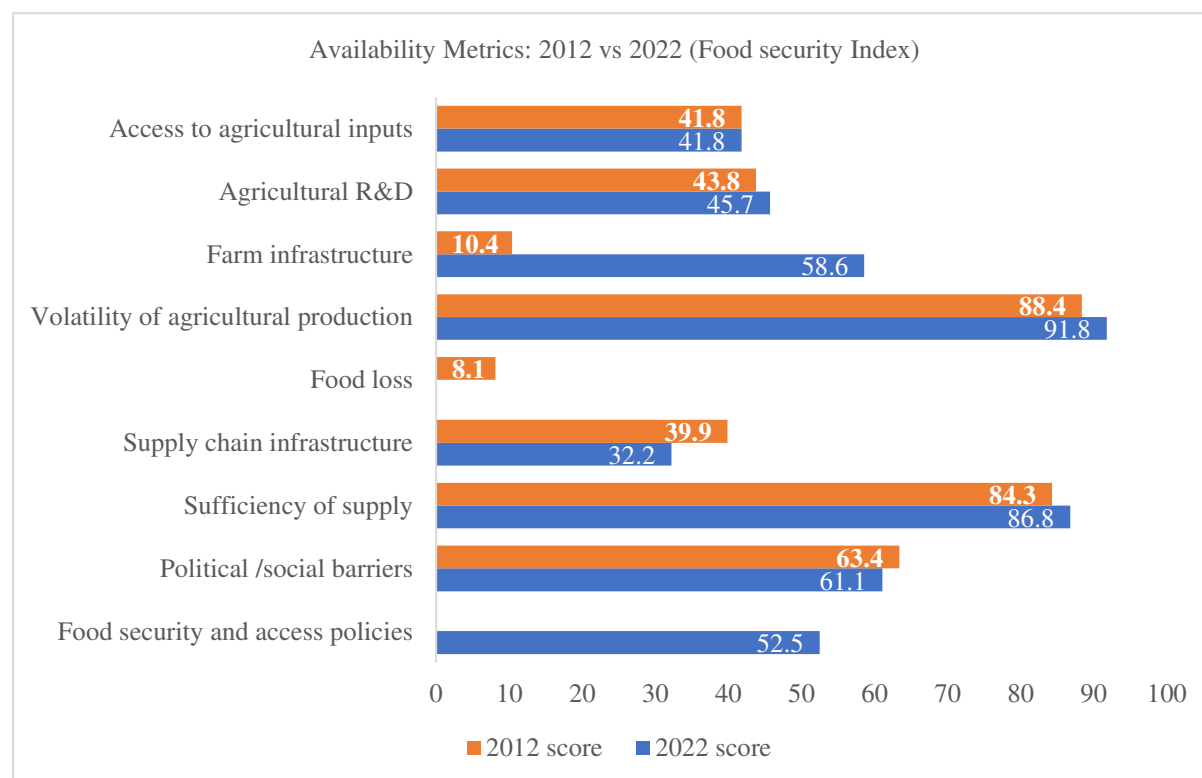
reduction. However, the slight decline of 6.1 points in "Food safety-net programmes" signals potential funding or implementation challenges in Ghana's social protection infrastructure.

Ghana's score of 45.4 in the "Inequality-adjusted income index" highlights persistent income inequality issues that affect food affordability. Though this score has improved by 4.8 points since 2012, it remains in the "weak" performance bracket, suggesting that income disparities continue to impact equitable access to food. The "Agricultural trade" indicator score of 61.9 (with a minimal decline of 0.7 points since 2012) reflects moderate performance in Ghana's agricultural trade policies and practices, which influence food prices and availability in domestic markets.

4.2.3 Availability

Ghana ranks 80th globally and 6th regionally in the Availability pillar, with a score of 52.4. This pillar has seen the most significant improvement over the past decade, increasing by 9.4 points since 2012. This positive trend suggests that Ghana has made substantial progress in enhancing the physical availability of food through improvements in production, storage, and distribution systems.

Figure 7: Ghana's food availability metrics (2012 vs 2022)



Source: Global Food Security Index, 2022

As shown in Figure 7, the most impressive improvements in this pillar have been in "Farm infrastructure" (58.6), which has increased by a remarkable 48.2 points since 2012. This dramatic improvement can be attributed to enhanced crop storage facilities, irrigation infrastructure, and access to market data and mobile banking. Implementing a food security strategy has also contributed to this positive trend. Ghana performs exceptionally well in certain aspects of food availability. The country scores in the "very good" bracket (80+) for two indicators: "Volatility of agricultural production" (91.8) and "Sufficiency of supply" (86.8). The high score in "Volatility of agricultural production" indicates that Ghana's agricultural sector has maintained relatively stable production levels, providing a reliable food supply. This indicator has improved by 3.4 points since 2012, suggesting increasingly stable agricultural output.

The "Sufficiency of supply" indicator has seen a substantial improvement, particularly since 2020, when the country scored only 35.4 points. The current score of 86.8 reflects reduced dependency on food aid and adequate food supply in the country. This improvement by 2.5 points since 2012 indicates positive developments in Ghana's food self-sufficiency. However,

significant challenges remain in the Availability pillar. The most alarming issue is food loss, for which Ghana scores 0 points, ranking last (113th) globally. This represents an all-time low and a decline of 8.1 points since 2012. This extremely poor performance suggests critical inefficiencies in Ghana's food supply chain, with substantial amounts of food being lost between harvest and consumption.

Other areas requiring attention include "Access to agricultural inputs" (41.8), "Agricultural research & development" (45.7), and "Supply chain infrastructure" (32.2). The score for "Supply chain infrastructure" has deteriorated by 7.7 points since 2012, indicating deteriorating conditions in transportation, storage, and distribution systems. The introduction of a food security strategy has contributed positively to the "Food security and access policy commitments" indicator, which has increased by 52.5 points. This significant improvement reflects enhanced policy attention to food security issues at the national level.

4.2.4 Quality and Safety

The Quality and Safety pillar represents one of Ghana's weaker areas, with a score of 50.5. Ghana ranks 94th globally and 14th regionally on this pillar. Despite the relatively low ranking, Figure 8 displays a rather modest improvement of 2.6 points since 2012, indicating some progress in enhancing the nutritional quality and safety of the food supply.

Figure 8: Ghana's food quality and safety metrics (2012 vs 2022)



Source: Global Food Security Index, 2022

Ghana's performance across the indicators in this pillar is highly variable. The country performs best in "Micronutrient availability," scoring 70.9 points and ranking joint 55th globally. This relatively strong performance can be attributed to a perfect score of 100 points in the sub-indicator for "Dietary availability of vitamin A," suggesting adequate supplies of this essential micronutrient in the Ghanaian diet.

However, significant weaknesses exist in other aspects of food quality and safety. The "Nutritional standards" indicator has experienced a dramatic decline, dropping from 57.1 points in 2021 to just 29.8 points in 2022. This substantial deterioration can be attributed to the absence of a national nutritional plan and strategy, for which Ghana scored 0 points. This deficiency highlights a critical gap in Ghana's policy framework for addressing nutrition issues.

Ghana's performance in "Dietary diversity" (46.1) has declined slightly by 2.3 points since 2012, suggesting limited variety in the typical Ghanaian diet. This is concerning as dietary diversity is essential for ensuring adequate intake of essential nutrients. The "Protein quality" indicator score of 42.9 indicates challenges in ensuring access to high-quality protein sources, although there has been a slight improvement of 0.5 points since 2012.

A notable positive trend is observed in the "Food safety" indicator, which has increased by 25.5 points since 2012 to reach a score of 63.3. This substantial improvement suggests enhanced regulations, monitoring, and enforcement of food safety standards in Ghana. However, there is still considerable room for improvement to reach "good" or "very good" performance levels. The mixed performance in this pillar suggests that while Ghana has made progress in specific aspects of food quality and safety, comprehensive policy approaches addressing nutritional standards and dietary diversity are lacking.

4.2.5 Sustainability and Adaptation

Sustainability and Adaptation represents Ghana's weakest pillar, with a score of 45.1. Ghana ranks 88th globally and 17th regionally on this pillar. Despite being the weakest component of Ghana's food security environment, this pillar has seen a slight improvement of 2.1 points since 2012, indicating some progress in addressing sustainability and climate adaptation issues.

The relatively poor performance in this pillar reflects significant challenges in ensuring the long-term sustainability of Ghana's food systems and building resilience to climate change and other environmental threats.

Table 5: Sustainability and adaptation metrics for Ghana (2012 vs 2022)

Metric	Score 2012	Score 2022	Change
Exposure	54.5	54.5	0.0
Water	27.6	27.6	0.0
Land	58.9	64.1	+5.2
Oceans, rivers and lakes	29.9	35.6	+5.7
Political commitment to adaptation	35.1	36.2	+1.1
Disaster risk management	53.9	53.4	-0.5

Source: Global Food Security Index 2022

The data reveals particularly weak performance in several critical areas. Water management represents a severe challenge, with Ghana scoring only 27.6 points in the "Water" indicator, placing it in the "very weak" performance bracket. This score has remained unchanged since 2012, suggesting persistent problems in water management and irrigation systems.

Similarly, the "Oceans, rivers and lakes" indicator shows a very low score of 35.6, although this has improved slightly by 5.7 points since 2012. These low scores highlight vulnerabilities

in Ghana's aquatic resources, which are crucial for both food production and environmental sustainability. Another critical weakness is in "Political commitment to adaptation," where Ghana scores only 36.2 points, ranking 88th globally. The low score in this indicator can be attributed to the lack of early-warning measures, insufficient implementation of climate-smart agriculture practices, and the absence of a comprehensive national agricultural adaptation policy. While this indicator has improved slightly by 1.3 points since 2012, the continued weak performance suggests inadequate policy attention to climate adaptation in the agricultural sector.

The "Exposure" indicator (54.5) and "Disaster risk management" indicator (53.4) both fall in the "weak" performance bracket, indicating vulnerabilities to climate-related shocks and limited capacity to manage disaster risks. The "Disaster risk management" score has seen only minimal improvement (0.5 points) since 2012, suggesting limited progress in building systems to anticipate, prepare for, and respond to disasters affecting food security.

The "Land" indicator presents a relatively better picture, with a score of 64.1 that has improved by 5.1 points since 2012. This moderate performance suggests some progress in sustainable land management practices, although there remains considerable room for improvement. The generally weak performance across this pillar highlights the need for Ghana to strengthen its commitment to sustainable food systems and climate adaptation. Given the increasing threats posed by climate change to agricultural production and food security, addressing these weaknesses should be a priority for policymakers. This has been summarized in Table 5

4.3. Key Factors Influencing Food Security and Nutritional Status in Ghana

Based on the Global Food Security Index data presented for Ghana (2022), the critical factors affecting food security and nutrition in the country has been presented

4.3.1. Escalating Food Costs and Weakened Safety Nets

The previous analysis showed that, the most alarming metric in the food availability score is the change in average food costs, scoring just 48.5 with a dramatic deterioration of -31.5 points. This indicates Ghanaians are facing substantially higher food prices compared to a decade ago, creating significant pressure on household budgets. Rising food inflation has been documented across Ghana, with the Ghana Statistical Service reporting consistently high food inflation rates

that outpace general inflation (Quaye, 2023). This aligns with broader regional trends where West African countries have experienced food price volatility exacerbated by climate shocks, currency depreciation, and supply chain disruptions (FAO, 2022).

Currency depreciation has significantly impacted food prices, with the Ghanaian cedi experiencing substantial volatility against major trading currencies. Between 2018 and 2022, the cedi depreciated by over 40% against the US dollar, directly increasing the cost of imported foods and agricultural inputs (Bank of Ghana, 2023). This currency weakness has been exacerbated by high inflation rates, with food inflation consistently outpacing general inflation reaching as high as 30.1% in 2022 (Ghana Statistical Service, 2023).

Global supply chain disruptions, initially triggered by the COVID-19 pandemic and later compounded by international conflicts, have further aggravated food price instability. The pandemic severely disrupted both domestic and international food supply chains, with border closures and movement restrictions creating artificial shortages and price spikes (Aday & Aday, 2020). These external shocks hit Ghana particularly hard due to its dependence on imported foods and agricultural inputs.

Rising production costs have created additional pressure on food prices. The costs of fertilizers, pesticides, and fuel have all increased substantially, with fertilizer prices more than doubling between 2020 and 2022 (Ministry of Food and Agriculture, 2023). These input cost increases directly translate to higher food prices in the market, as smallholder farmers who produce approximately 80% of Ghana's domestic food operate on thin margins and must pass cost increases to consumers (Opoku Mensah et al., 2025).

Simultaneously, safety net programs designed to protect vulnerable populations have weakened with a decline of 6.1. The coverage and effectiveness of Ghana's flagship social protection programs, including the Livelihood Empowerment Against Poverty (LEAP) program, have not expanded proportionally to need. LEAP reaches only about 30% of eligible impoverished households, leaving significant gaps in coverage (World Bank, 2022). Funding constraints have limited both the scope and impact of these programs, with per-household transfer values failing to keep pace with inflation, thereby eroding their purchasing power and protective effect (Okoth, 2021).

During economic downturns, informal safety nets like community support systems and family networks also face strain as more households experience financial pressure simultaneously. Research by Olawuyi (2019) demonstrated that traditional reciprocity networks that historically helped buffer food insecurity have become less effective as economic pressures have become more widespread.

4.3.2 Poor Supply Chain and High Food Loss

The availability pillar presents a mixed picture with an overall score of 52.4 and improvement of +9.4 points. However, looking deeper reveals critical infrastructure and supply chain weaknesses that undermine Ghana's food security.

Food loss (0.0, deteriorated by -8.1) and supply chain infrastructure (32.2, deteriorated by -7.7) are the most concerning metrics. Both fall into the "very weak" category and have worsened substantially since 2012. These scores indicate failures in systemic post-harvest handling, storage, and distribution systems. According to Fukah et al. (2024), post-harvest losses in Ghana range from 20-50% for various staple crops, representing not only lost nutrition but also wasted resources, labor, and environmental inputs. The Ghana Food Loss Index compiled by the Ministry of Food and Agriculture corroborates these findings, estimating annual losses valued at over \$700 million (MOFA, 2023).

Transportation inefficiencies further compromise food distribution networks. The poor condition of rural roads, particularly in major food-producing regions, creates significant bottlenecks in moving produce from farms to markets. During rainy seasons, many agricultural communities become effectively isolated due to impassable roads (Morgan, 2019). The World Bank's Rural Accessibility Index indicates that only 61% of Ghana's rural population lives within 2 kilometers of an all-season road, significantly below the recommended 80% threshold for effective agricultural commercialization (World Bank, 2021).

Limited cold chain infrastructure particularly affects high-value, nutritious perishables such as fruits, vegetables, dairy, fish, and meat. Less than 5% of perishable produce in Ghana passes through a proper cold chain, compared to over 85% in developed food systems (IFPRI, 2022). This gap not only contributes to physical losses but also to nutritional deterioration, as micronutrients degrade rapidly under ambient tropical conditions. The processing sector

suffers from capacity constraints and technological limitations. Small and medium-scale processors often operate with outdated equipment, intermittent electricity, and limited technical knowledge. A survey by the Food Research Institute found that approximately 70% of food processing enterprises in Ghana use technologies that are at least 15 years behind current global standards (CSIR-FRI, 2022).

Despite these persistent challenges, farm infrastructure has seen remarkable improvement (+48.2) to a moderate score of 58.6. Irrigation systems have expanded, albeit modestly, with approximately 0.89% of agricultural land now under irrigation compared to 0.5% a decade ago (MOFA, 2023). Farm mechanization has increased, with the tractor-to-farm ratio improving from 1:1,500 to 1:800 over the past decade (IFPRI, 2023). These production-focused improvements have not been matched by corresponding investments in post-harvest infrastructure, creating a bottleneck where increased production meets inadequate handling and distribution systems.

4.3.3 Inadequate Nutritional Standards and Low Dietary Diversity

The quality and safety pillar scores 50.5 with improvement of +2.6 points, but this modest overall improvement masks serious nutritional deficiencies. Nutritional standards are extremely concerning, scoring just 29.8 with significant deterioration (-11.3) since 2012. This places Ghana's nutritional standards in the "very weak" category, suggesting inadequate regulatory frameworks, monitoring systems, or enforcement mechanisms to ensure nutritional quality of the food supply. Jaffee et al. (2018) found that Ghana's food regulatory system faces significant challenges in monitoring and enforcing standards, particularly for locally produced and processed foods that comprise a significant portion of the diet for lower-income households.

Dietary diversity also presents concerns, scoring 46.1 with deterioration (-2.3). This indicates that despite adequate calories, many Ghanaians may not be consuming sufficiently varied diets to meet micronutrient needs. Research by the Ghana Health Service has consistently identified monotonous diets heavily reliant on starchy staples as a key contributor to malnutrition, particularly among children and women of reproductive age (Ghana Health Service, 2022). A study by Vercillo et al. (2025) found that even in regions with adequate food quantity, dietary

quality and diversity remained poor, with over 60% of households failing to meet minimum dietary diversity standards.

Micronutrient availability scores relatively well (70.9) but has slightly deteriorated (-0.2), suggesting challenges in maintaining consistent access to essential vitamins and minerals. Protein quality registers a weak score (42.9) with minimal improvement (+0.5), indicating continued challenges in access to quality protein sources. This aligns with findings from the Ghana Micronutrient Survey, which identified persistent deficiencies in iron, vitamin A, and zinc despite various intervention programs (University of Ghana, 2022).

The bright spot in this category is food safety, scoring 63.3 with substantial improvement (+25.5). This suggests significant strides in preventing foodborne illnesses and contamination, likely through improved handling practices, inspection systems, and regulations. However, this contrasts with the poor nutritional standards score, indicating that while food may be safer from pathogens, its nutritional quality remains inadequate.

4.3.4 Weak Water and Disaster Risk Management

The sustainability and adaptation pillar scores a weak 45.1 but shows improvement (+2.1), with significant variations across individual metrics that highlight vulnerability to environmental stressors. Water resources represent the most critical vulnerability, scoring just 27.6 (unchanged) and falling into the "very weak" category. This indicates severe challenges in water management, irrigation infrastructure, and resilience to water stress that directly impact agricultural productivity and food security. Research by the Water Research Institute of Ghana has documented declining water availability for agriculture due to combination of climate change, watershed degradation, and competing uses from industrial and urban sectors (Water Research Institute, 2023). Kyei-Mensah et al. (2019) found that less than 2% of Ghana's cultivated land has reliable irrigation, leaving most production vulnerable to rainfall variability.

Watershed degradation has compromised water availability and quality throughout agricultural regions. Deforestation, mining activities (particularly illegal small-scale gold mining or "galamsey"), and urban expansion have damaged critical watersheds. The Water Resources Commission reports that approximately 60% of Ghana's watershed areas show significant degradation, with direct impacts on agricultural water availability (Water Resources

Commission, 2023). The contamination of water bodies by heavy metals from mining activities has rendered substantial water resources unusable for irrigation or livestock, particularly in the Western, Ashanti, and Eastern regions.

Irrigation infrastructure remains severely limited despite its potential to buffer climate variability. Ghana has developed less than 20% of its irrigation potential, with just 0.89% of agricultural land under irrigation (MOFA, 2023). This limited coverage leaves the vast majority of production vulnerable to rainfall fluctuations. The Ghana Irrigation Development Authority has faced persistent challenges in expanding irrigation infrastructure, including financing constraints, land tenure complexities, and maintenance difficulties (GIDA, 2022).

Groundwater management faces significant challenges, with unregulated extraction in some areas leading to declining water tables and threatening long-term sustainability. Research by the Water Research Institute identified concerning groundwater depletion trends in key agricultural regions, with water tables dropping by an average of 0.5 meters annually in intensively farmed areas (WRI, 2023).

Disaster risk management systems demonstrate several critical weaknesses that undermine food system resilience. Early warning systems for climate-related disasters remain inadequately developed, particularly at the local level where they're most needed. The National Disaster Management Organization's capacity for proactive disaster risk reduction remains limited by resource constraints, with its budget representing less than 0.5% of the national budget despite the increasing frequency of climate-related disasters (NADMO, 2022).

Community-level disaster preparedness shows significant gaps, with a survey of agricultural communities finding that only 32% had disaster preparedness plans or had received related training (CARE Ghana, 2022). This preparedness deficit leaves communities vulnerable when disasters occur, prolonging recovery timeframes and deepening food security impacts.

Post-disaster recovery mechanisms also demonstrate weaknesses, with limited insurance coverage and slow-moving government relief efforts. Agricultural insurance covers less than 1% of Ghana's farmers, leaving the vast majority without financial protection against climate shocks (Ghana Agricultural Insurance Pool, 2023). Government relief efforts following disasters often face delays and coordination challenges, extending the recovery timeframe for

affected food systems (Agyemang & Kwofie, 2021). Climate adaptation capacity reflects significant constraints at both institutional and farm levels. Ghana's climate adaptation policies exhibit implementation gaps, with a disconnection between national-level planning and local-level action. Smallholder farmers face multiple barriers to adopting climate-smart agricultural practices, including knowledge gaps, financial constraints, and limited access to appropriate technologies (IFPRI, 2023).

4.4 Existing Government Policies, Interventions, and Progress Made in Addressing Food Security Challenges in Ghana

This section presents the government policies and interventions implemented to address food security challenges, evaluates their efficacy, and assesses the progress made toward ensuring food security in Ghana.

4.4.1 Planting for Food and Jobs (PFJ)

Launched in 2017, the Planting for Food and Jobs (PFJ) program represents one of Ghana's most ambitious agricultural interventions aimed at addressing food security challenges. The program operates on five pillars: provision of improved seeds, supply of fertilizers, agricultural extension services, marketing, and e-agriculture. The food policy in its first year of implementation focused on the production of maize, rice sorghum, soy bean and some selected vegetables that is onion tomato and chili pepper (Pauw, 2022). Through subsidized inputs and technical support, PFJ aims to boost crop production, create employment opportunities, and reduce food imports.

Table 6: selected crop production in Ghana from 2012 to 2021('000 metric tons)

Crop	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
Maize	1,950	1,764	1,769	1,692	1,722	2,011	2,306	2,911	3,071	3,203
Rice(paddy & milled)	813	963	1,021	1,084	1,163	1,221	1,300	1,563	1,624	1,863
Sorghum	280	257	259	263	230	278	316	347	356	324
Soyabean	152	139	141	142	143	170	177	193	209	218

Source: MoFA, 2022

Shortly after its implementation PFJ produced recognizable results. Yields for maize and rice and soy beans increased from 1.8mt/ha to 3,4 mt/ha; 2,7mt/ha to 4mt/ha and 1mt/ha to 3mt/ha

respectively. In addition, the Agricultural Gross Domestic Product increased from 2.9% in 2016 to 6.1% in 2017. (Agbolosoo& Hutagaol,2024). Though a well needed policy it was not without some failures. The issue of low quantity inputs like seeds and fertilizers posed a hindrance to farmers. Also funding for the project was not evenly distributed among the pillars as only 13% of the beneficiaries had access to marketing and e-agriculture(which covers distribution of information on weather and input prices) (Pauw, 2022).

4.4.2 Ghana School Feeding Programme (GSFP)

The Ghana School Feeding Programme, initiated in 2005, exemplifies a direct intervention aimed at improving food utilization while simultaneously boosting educational outcomes and stimulating local agricultural markets (Abizari et al., 2014)By providing one hot, nutritious meal per day to schoolchildren in deprived communities, the program addresses immediate hunger needs while creating demand for locally produced food items.

Table 7: Ghana school feeding programme: impacts and challenges, 2018-2022

Indicator	2018	2019	2020	2021	2022
Number of beneficiary schools	8,683	9,178	9,572	10,832	11,050
Number of beneficiary children	2,824,738	3,043,226	3,240,150	3,448,065	3,562,370
Budget allocation (GHC million)	456	490	520	642	686
Disbursement rate (%)	76%	81%	62%	74%	79%
Local food sourcing (%)	59%	63%	58%	65%	65%

Source: Ghana School Feeding Secretariat, 2023

Evidence indicates positive impacts on school enrollment, attendance, and retention rates, with participating schools reporting an average 20% increase in enrollment (Mensah, 2021). However, analysis by Sulemana et al. (2022) revealed operational inefficiencies, including irregular fund disbursement and inconsistent food quality standards. Additionally, while the program aims to source 80% of food locally, actual local procurement averaged only 62% between 2018 and 2022, limiting the program's potential to stimulate local agricultural production (Ghana School Feeding Secretariat, 2023).

4.4.3 One Village, One Dam Initiative

The One Village, One Dam (1V1D) initiative, launched in 2017, aims to enhance year-round agricultural production through improved water management, particularly in northern Ghana,

where rainfall patterns are less reliable (MoFA, 2022). The initiative plans to construct small-scale dams in every village across the northern regions to facilitate dry-season farming and livestock watering.

Table 8: one village, one dam implementation status (2023)

Region	Planned Dams	Completed Dams	Functional Dams	Quality Concerns
Northern	165	114	72	42
Upper East	112	73	48	25
Upper West	114	78	39	39
North East	94	52	31	21
Savannah	85	45	23	22
Total	570	362	213	149
Percentage	100%	63.5%	37.4%	41.2%

Source: Ghana Irrigation Development Authority (2023)

Implementation analysis reveals significant discrepancies between policy objectives and outcomes. By 2023, approximately 65% of planned dams had been constructed, but quality concerns and functionality issues affected about 40% of completed projects (Ghana Irrigation Development Authority, 2023). Hydrological studies conducted by Nyamekye et al. (2021) found that several dams had inadequate water retention capacity or were constructed in unsuitable locations, limiting their effectiveness for agriculture. Community participation in planning and maintenance also varied considerably, affecting long-term sustainability (Adams et al., 2024).

4.4.4 National Food Buffer Stock Company (NAFCO)

Established in 2010, the National Food Buffer Stock Company (NAFCO) aims to stabilize food prices, minimize post-harvest losses, and ensure food availability throughout the year by purchasing excess production during harvest seasons (NAFCO, 2022). This intervention addresses the stability dimension of food security by moderating price fluctuations and ensuring continuous supply.

Analysis by Adom et al. (2022) found that NAFCO's operations contributed to an average 15% reduction in seasonal price volatility for key staples between 2015 and 2021. However, limited storage infrastructure and financial constraints restricted the company's ability to purchase

sufficient quantities during peak harvest seasons, reducing its market stabilization potential (Teye and Torvikey, 2021). Moreover, centralized procurement and distribution systems created inefficiencies, particularly in serving remote communities (Mensah and Brümmer, 2022).

4.5 Ghana's Progress Towards Achieving SDG 1 (No Poverty), SDG 2 (Zero Hunger), and SDG 3 (Good Health and Well-being) by 2030

Ghana has made strides toward achieving Sustainable Development Goals (SDGs) 1 (No Poverty), 2 (Zero Hunger), and 3 (Good Health and Well-being). However, significant challenges persist, requiring acceleration in implementation efforts. Table 5 is an analytical overview of Ghana's progress with data presented in table format.

SDG	Key Targets	Current Status	Key Achievements	Challenges	Achievement Level
SDG 1: No Poverty	• Eradicate extreme poverty (living on less than \$1.90/day) • Reduce poverty by half according to national definitions • Implement social protection systems • Equal rights to economic resources	• Extreme poverty rate: ~9% globally • Social protection coverage: ~45% of global population	• Reduction in poverty rates through social protection programs • Improved access to basic amenities like water and electricity	• Financing gap of \$43 billion annually • COVID-19 disruptions increasing poverty levels	Off-track <i>Progress made but unlikely to achieve 2030 targets</i>
SDG 2: Zero Hunger	• End hunger and ensure food access • End all forms of malnutrition • Double agricultural productivity of small-scale producers • Ensure sustainable food systems	• ~690 million people undernourished globally • Child stunting: ~22% • Child wasting: ~7%	• Reduction in stunting and overweight among children under five • Initiatives like "Planting for Food and Jobs" promoting sustainable agriculture	• Malnutrition and wasting among children persist • Rural food security stagnates	Partially on track <i>Some targets achievable, others unlikely</i>
SDG 3: Good Health and Well-being	• Reduce maternal mortality to <70 per 100,000 live births • End preventable deaths of newborns and children under 5 • End epidemics of AIDS, TB, malaria • Universal health coverage	• Maternal mortality: ~211 per 100,000 live births globally • Under-5 mortality: ~39 per 1,000 live births • UHC service coverage: ~66%	• Maternal mortality rates reduced • Improved access to healthcare services	• Stagnation in combating communicable diseases • Challenges in sub-national implementation	Mixed progress <i>Some indicators improving, others stagnating</i>

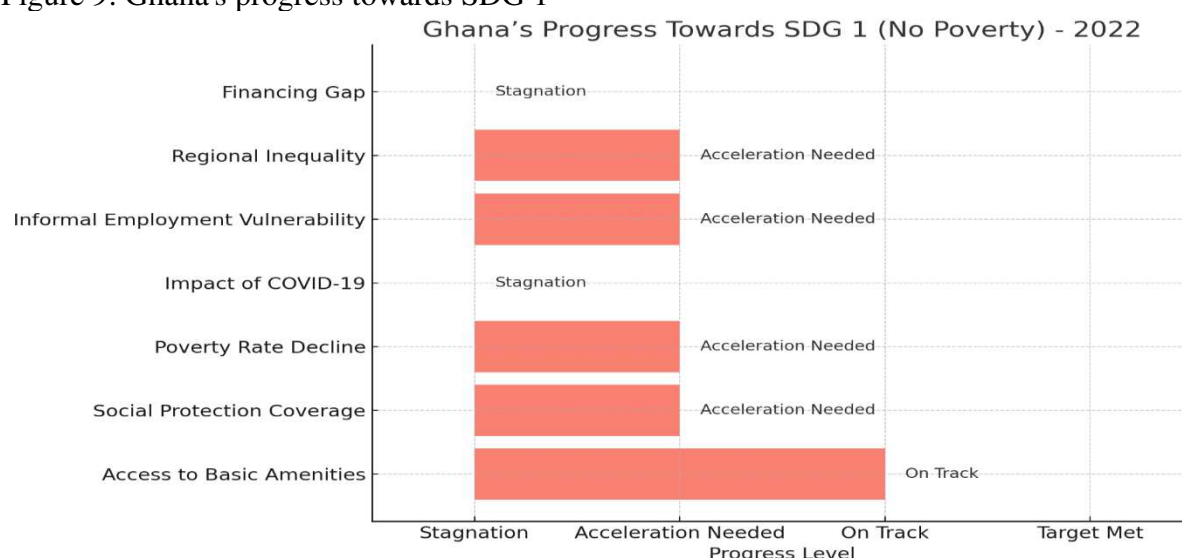
Table 9: SDGs 1;2;3 Progress

Source: (National Development Planning Commission, 2022)

4.5.1 SDG 1: No Poverty

Ghana's progress towards achieving SDG1 (No Poverty) has seen positive developments and persistent challenges. The country has made commendable strides through various social protection and poverty alleviation programs. Initiatives such as the Livelihood Empowerment Against Poverty (LEAP), the National Health Insurance Scheme (NHIS), and Free Senior High School education have contributed to reducing multidimensional poverty and improving access to essential services among vulnerable populations. These interventions have improved access to basic amenities such as electricity, clean water, and education, which are critical components in the fight against poverty (NDPC, 2022).

Figure 9: Ghana's progress towards SDG 1



Source: (National Development Planning Commission, 2022)

Despite these efforts, the goal of eradicating extreme poverty by 2030 remains off track. The COVID-19 pandemic significantly disrupted Ghana's economic progress, reversing some of the gains made in poverty reduction. The pandemic exacerbated income inequalities, increased unemployment rates, and weakened household purchasing power, particularly among informal sector workers who form a large portion of Ghana's workforce (Issahaku & Musah Abu, 2020). Additionally, rising inflation and the depreciation of the local currency have affected the cost of living and pushed more households below the poverty line (NDPC, 2022).

One of the critical barriers to achieving SDG 1 is the high dependency on informal employment, which lacks job security and social protection. A large segment of Ghana's population continues to rely on subsistence farming and informal trade, which are highly vulnerable to external shocks, climate variability, and market fluctuations. Furthermore, regional disparities in poverty remain stark, with northern regions of the country consistently recording higher poverty rates than the south. These inequalities underscore the need for geographically targeted interventions and inclusive growth strategies that prioritize marginalized and underserved communities (NDPC, 2022).

The financing gap also poses a significant threat to the achievement of SDG 1. According to the report, Ghana requires an estimated \$43 billion annually to meet the SDG targets, with poverty reduction efforts constituting a large portion of this financial need. The reliance on external aid and debt-financed development projects has led to sustainability concerns, especially given the country's recent fiscal constraints. To overcome these barriers, Ghana must mobilize domestic resources more effectively, enhance revenue collection, and eliminate inefficiencies in public spending (NDPC, 2022).

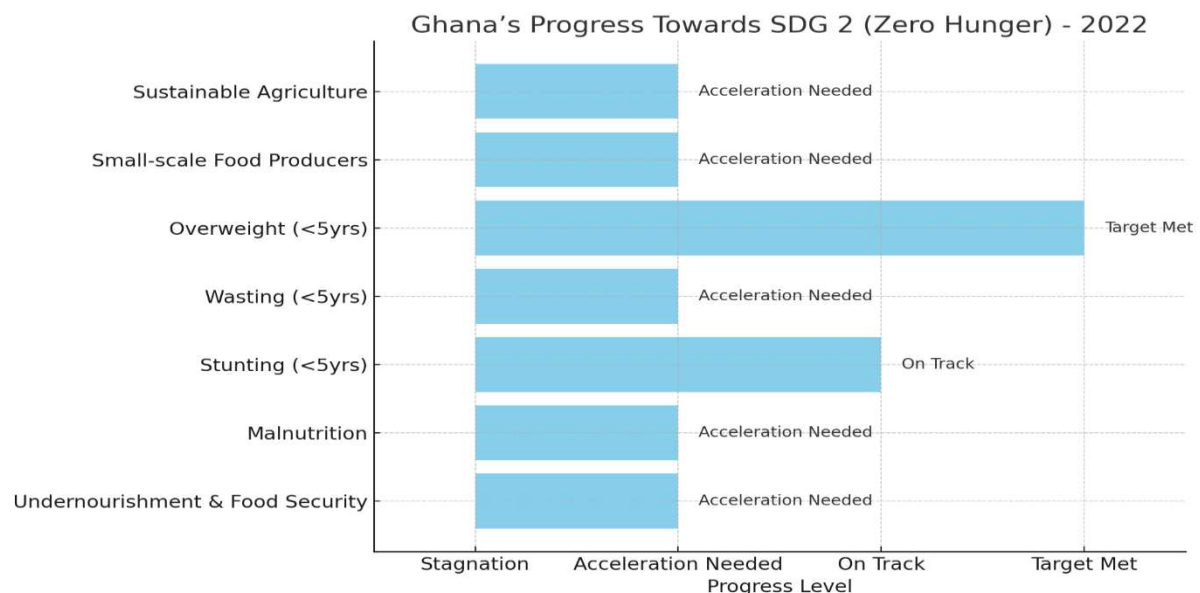
To accelerate progress toward eradicating poverty, Ghana must expand social protection coverage, particularly for informal workers, and invest in resilient and inclusive economic growth. Strengthening rural infrastructure, improving access to credit for small-scale entrepreneurs, and promoting value addition in agriculture can provide sustainable livelihoods and reduce poverty. Furthermore, integrating poverty data into planning and policy implementation can ensure that resources are directed where they are most needed, thereby promoting equity and long-term impact. Without addressing these structural challenges, achieving SDG 1 by 2030 may remain elusive (NDPC, 2022).

4.5.2 SDG 2: Zero Hunger

As indicated in figure 9, Ghana's progress toward achieving Sustainable Development Goal 2 (Zero Hunger) presents a mixed picture. While some critical nutritional targets have been met, others still require significant acceleration. One of the notable achievements is the successful reduction of overweight among children under five years, which stood at 2.9% as of 2020. This indicates that Ghana has met the global target for this indicator. Similarly, the country is on

track in its efforts to reduce stunting among children under five, with females and urban populations showing better outcomes compared to males and rural dwellers. For instance, regions like Greater Accra are on track in addressing stunting, whereas the Northern region continues to face stagnation in this area.

Figure 10: Ghana's progress towards SDG 2



Source: National Development Planning Commission, 2022

Despite these gains, Ghana faces substantial challenges in other nutritional areas. Undernourishment and food security remain pressing issues, especially in rural communities and northern regions such as Upper West, Upper East, and Northern Ghana. These areas are experiencing stagnation and require intensified efforts to improve food access and nutritional outcomes. Malnutrition and wasting among children under five also demand urgent attention, as progress has been slow, and targets are far from being met. In addition, while there have been national programs aimed at empowering small-scale food producers and enhancing sustainable agriculture, these sectors still require accelerated investment and support to achieve SDG 2 targets (kolog et al, 20223).

Several structural challenges continue to impede Ghana's ability to fully realize the Zero Hunger goal. Chief among these is the limited mechanization of the agricultural sector, which relies heavily on traditional farming practices. Poor financing and investments in agriculture

further exacerbate the issue, alongside high production and food processing costs due to expensive inputs like seeds and fertilizers. Additionally, factors such as climate change, environmental degradation, rapid population growth, poor road networks, and land encroachment by nomadic farmers contribute to food insecurity and post-harvest losses (Choudhary et al,2015). The COVID-19 pandemic has also intensified food inflation and disrupted food supply chains, making it more difficult for vulnerable populations to access nutritious food.

Although Ghana has introduced notable policies and programs—such as Planting for Food and Jobs, and school-based nutrition programs their impact is uneven across regions. Urban communities and selected regions like Ashanti and Brong Ahafo are on track to achieving food security, but others lag behind. Furthermore, Ghana faces significant data gaps in tracking progress on genetic agricultural resources, food price anomalies, and agricultural export subsidies, limiting effective monitoring and response.

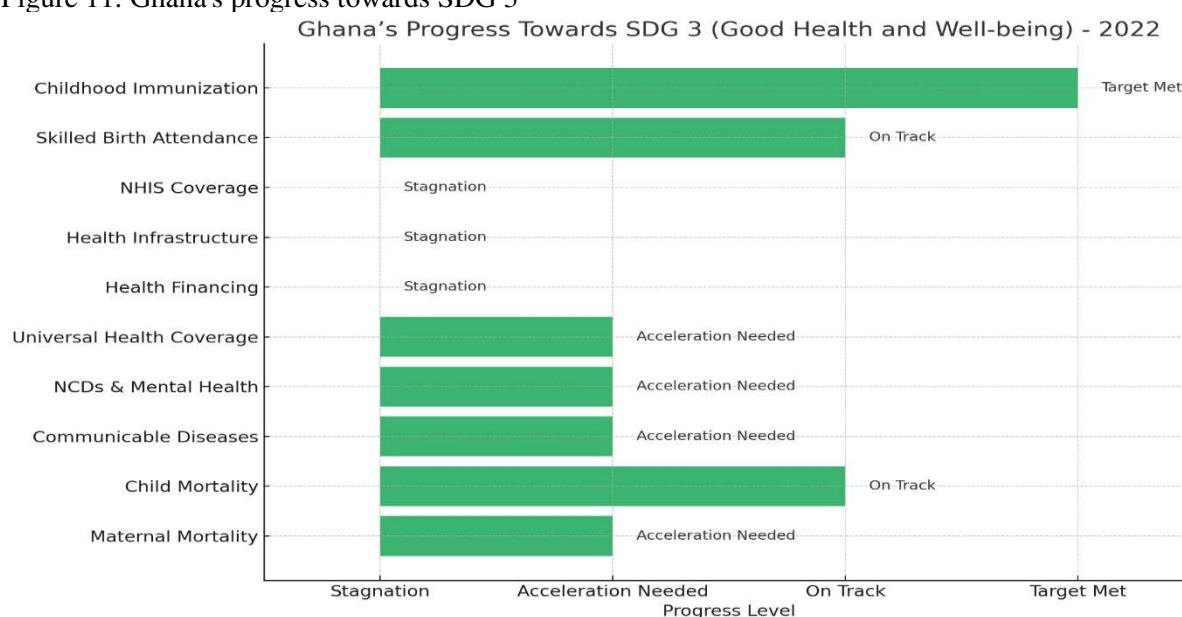
To achieve SDG 2 by 2030, Ghana must strengthen agricultural investments, scale up mechanization, improve rural infrastructure, and enhance the nutritional status of children, especially in underserved regions. Equally important is building a robust data collection and monitoring system to support evidence-based interventions and equitable progress across all regions and population groups.

4.5.3 SDG 3: Good Health and Well-being

Ghana's progress towards Sustainable Development Goal 3 (Good Health and Well-being) reflects a combination of notable improvements and significant gaps. The country has made commendable strides in reducing maternal and child mortality. Maternal mortality dropped from 580 deaths per 100,000 live births in 2007 to 301 in 2021, although further acceleration is needed to meet the global target of 70 per 100,000 by 2030 (NDPC, 2022). Similarly, child mortality rates have also improved over the years due to expanded access to healthcare services, especially skilled birth attendance and childhood immunization. The expansion of the Community-Based Health Planning and Services (CHPS) compounds and the use of medical drones for vaccine delivery in hard-to-reach areas have contributed to these gains (NDPC, 2022).

Moreover, Ghana has demonstrated strong commitment in combating communicable diseases such as HIV, tuberculosis, and malaria. However, while infection rates have shown some decline, the progress remains slow, and consistent investment is needed. The incidence of HIV among adolescents and children is still a concern, and antiretroviral therapy coverage is not yet optimal. Non-communicable diseases (NCDs) and mental health are also emerging areas that require significant attention. While Ghana has taken steps to integrate mental health into primary healthcare and promote awareness, the lack of data and underfunding remain critical challenges (NDPC, 2022). Research has shown that negative factors like food insecurity affect the mental health and development of children (Kusi- Mensah et al, 2022).

Figure 11: Ghana's progress towards SDG 3



Source: National Development Planning Commission, 2022

Universal Health Coverage (UHC) is a central goal under SDG 3, and while progress has been made, challenges persist. The National Health Insurance Scheme (NHIS), which is intended to provide financial protection for healthcare access, saw a decline in population coverage from 38.4% in 2013 to 35.8% in 2018. This is attributed to poor financing and administrative inefficiencies. Additionally, geographical disparities in access to health services, especially in rural areas, and the inequitable distribution of healthcare professionals continue to impede equitable healthcare delivery (NDPC, 2022).

Furthermore, there is insufficient progress in health system financing and infrastructure. The health sector remains underfunded, and although core health expenditure increased from

GHC3,055.98 million in 2018 to GHC4,402.7 million in 2021, this investment is still not enough to meet the growing population's needs (NDPC, 2022). Access to reproductive health services, prevention of road traffic accidents, and policies on substance abuse are among areas where data is lacking, making it difficult to track progress and design effective interventions.

Ghana's subnational performance also shows disparities. For instance, childhood immunization targets have been met in 15 out of 16 administrative regions, and urban areas have met targets for skilled birth attendance, whereas rural areas are still catching up. Additionally, adolescent birth rates have declined, reflecting some success in reproductive health education and access (NDPC, 2022).

In summary, while Ghana has achieved progress in maternal and child health and made efforts in disease control, systemic issues such as underfunding, poor data tracking, workforce imbalances, and service delivery gaps persist. Addressing these challenges through targeted investments, health system reforms, and improved data systems is essential to fully achieving SDG 3 by 2030.

5 CONCLUSION AND RECOMMENDATIONS

5.1. Conclusions

This study provides a comprehensive analysis of nutritional security and nutritional status in Ghana. This is directed towards four main goals; examining current trends in nutrition security with focus on dimensions such as availability, affordability, quality, security, sustainability, and adaptation; identifying important factors influencing nutritional safety and nutrition; evaluating government policies and interventions effectiveness ; as well as evaluating Ghana's progress in achieving SDGS 1 (no poverty), 2 (zero hunger), and 3 (good health and happiness) by 2030.

While there has been incremental progress in some areas such as farm infrastructure and poverty reduction, food affordability has significantly declined due to rising food prices, currency depreciation, and weakened social safety nets. The affordability index, despite being Ghana's strongest scoring pillar, declined by 4.4 points since 2012, reflecting worsening food access for vulnerable populations. Food availability improved marginally, with notable gains in agricultural production stability and sufficiency of supply. On quality and safety, the country's nutritional standards showed a dramatic drop, compounded by the absence of a national nutrition policy. Similarly, although micronutrient availability scored moderately well, issues of dietary diversity and poor protein quality remain persistent. Ghana's performance in sustainability and adaptation which is its weakest dimension, reflects limited political commitment to climate-smart agriculture, inadequate water resource management, and weak disaster preparedness frameworks.

In terms of nutrition, the double burden of malnutrition has become increasingly evident. The study documents coexisting challenges of undernutrition (e.g., stunting and wasting in children) and rising overnutrition (e.g., obesity and diet-related non-communicable diseases in adults), particularly among urban populations. Micronutrient deficiencies such as anaemia and vitamin A deficiency remain prevalent, especially in northern Ghana and rural areas, where food insecurity is also most acute.

5.2. Recommendations

Based on the results of this study, the following specific recommendations are proposed to enhance food security and nutritional status in Ghana:

1. **Reduce Post-Harvest Food Loss Through Infrastructure Investment:**

Ghana's score of 0.0 in the GFSI food loss indicator highlights severe inefficiencies in post-harvest handling. The government should prioritize investments in cold storage, transportation, and processing facilities, especially in major food-producing regions, to minimize losses and improve food availability.

2. **Develop and Implement a National Nutrition Strategy:**

The absence of a national nutrition plan resulted in a drastic decline in Ghana's nutritional standards score. A multisectoral nutrition strategy—integrated with health, education, and agriculture sectors—is urgently needed to address undernutrition, micronutrient deficiencies, and rising obesity.

3. **Reinforce and Expand Social Protection Programs:**

Food affordability has declined significantly due to inflation and currency depreciation. Programs like the Livelihood Empowerment Against Poverty (LEAP) should be scaled up, with benefit amounts adjusted for inflation to maintain real value and protect vulnerable populations from food insecurity.

4. **Strengthen Climate Adaptation in Agriculture:**

Ghana's weak score in sustainability and adaptation reflects poor resilience to environmental shocks. Policies should support climate-smart agriculture, early warning systems, and water management interventions to enhance food system stability in the face of climate change.

5. **Target Regional Disparities with Localized Interventions:**

With northern Ghana facing disproportionately higher food insecurity and malnutrition rates, targeted interventions such as dry-season irrigation, community nutrition programs, and improved rural infrastructure should be deployed to bridge the north-south food security gap.

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